

ThinkShield Web application

User role capabilities and how to update the Device key

The Lenovo logo is positioned in the top right corner of the slide. It consists of the word "Lenovo" in white, sans-serif font, oriented vertically. The text is set against a rectangular background with a vertical color gradient transitioning from green at the top to blue at the bottom.

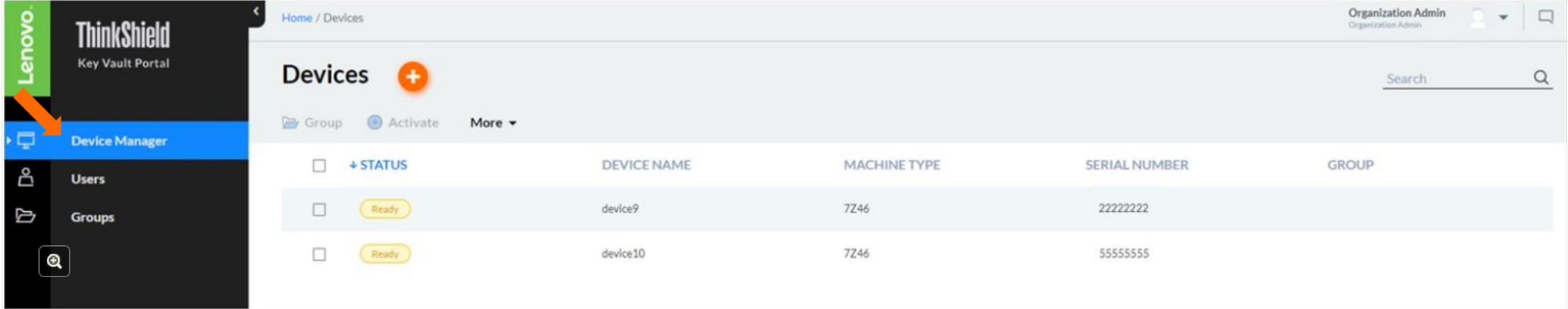
Lenovo

ThinkShield Web application user roles and capabilities

Capabilities / Roles	Has a Lenovo ID	Base user	Edge user	Maintenance user	Org Admin
Log in to / log out of the ThinkShield Mobile application		V	V	V	V
Edit my profile		V	V	V	V
Access to an organization		V	V	V	V
Create an organization	V				

User roles and the ThinkShield Web application Device Manager tab

Device Manager / Roles	Edge user	Maintenance user	Org Admin
Claiming / unclaiming a device	V	V	V
Automatic Activation	V	V	V
Manual Activation	V	V	V
Re-sync	V	V	V
Edit device name	V	V	V
Update Device key		V	
Transfer			V
Import / Export devices			V
Bulk claim		V	V



The screenshot displays the ThinkShield Key Vault Portal interface. On the left, a navigation sidebar shows the 'Device Manager' tab selected, indicated by a blue bar and an orange arrow. The main content area is titled 'Devices' and features a table with columns: STATUS, DEVICE NAME, MACHINE TYPE, SERIAL NUMBER, and GROUP. The table lists two devices, 'device9' and 'device10', both with a 'Ready' status. The top right of the interface shows the user role as 'Organization Admin'.

STATUS	DEVICE NAME	MACHINE TYPE	SERIAL NUMBER	GROUP
Ready	device9	7Z46	22222222	
Ready	device10	7Z46	55555555	

User roles and the ThinkShield Web application Users tab

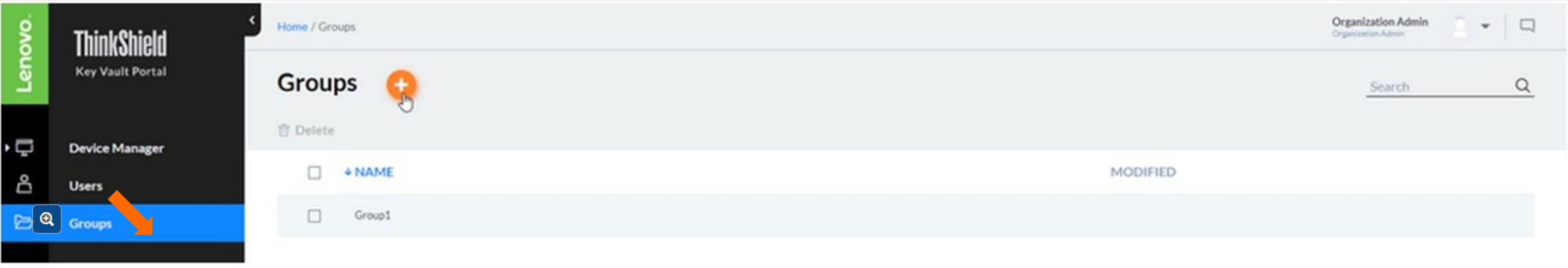
Users / Roles	Edge user	Maintenance user	Org Admin
Invite a User			V
Remove a User			V
Import / Export Users			V
Bulk Invite Users			V

The screenshot shows the ThinkShield Web application interface. On the left is a dark sidebar with the Lenovo logo and 'ThinkShield Key Vault Portal' text. Below this are menu items: 'Device Manager', 'Users' (highlighted in blue with an orange arrow pointing to it), and 'Groups'. The main content area has a light blue header with 'Home / Users' and user information 'Admin Admin LCP Admin'. Below the header is a 'Users' section with a '+ Add' button and a 'Delete' button (highlighted with an orange arrow). A table lists users with columns: NAME, ROLE, EMAIL, STATUS, and GROUP. One user, 'John Doe', is listed with the role 'Organization Admin', email 'newuser@test.com', and status 'Invited'. A search bar is located on the right side of the table.

NAME	ROLE	EMAIL	STATUS	GROUP
John Doe	Organization Admin	newuser@test.com	Invited	

User roles and the ThinkShield Web application Groups tab

Groups / Roles	Edge user	Maintenance user	Org Admin
Create a group		V	V
Delete a group		V	V
Group devices / Remove devices from group		V	V



ThinkShield Web application – Update Key

After replacing a system board, Maintenance users can work through the following procedure to update a Device key using the ThinkShield Web application:

Steps	Web interface	IPMI
1	Replace the system board (update the firmware and VPD), and then prepare a laptop	
2	Open a browser and log in to ThinkShield Portal	
3		Connect the laptop to the SE350 XCC network
4		Get the Device key with IPMI
5	Update the key with the Device key obtained using IPMI (If you need to claim the device, use the old activation code from the flyer or asset tag)	
6	Check whether the device is Standard or Security Pack	
7		Issue an IPMI command to change the personality only if it should be Security Pack

Step



ThinkShield Web application – Update Key

Select a device to open the device tray. Select **Update Key**, and then click **Save**.

The screenshot displays the ThinkShield Web application interface. On the left, a sidebar shows the navigation menu with 'Device Manager' selected. The main content area shows a table of devices. A blue arrow labeled 'Device tray' points to the right-hand panel, which is highlighted with a red border. This panel shows the details for a selected device, 'device3', and a list of actions: 'Activate', 'Manually Activate', 'Update Key' (highlighted with a red arrow), 'Restore Key Backup', 'Re-sync', and 'Unclaim'. At the bottom right of the red-bordered panel are 'Cancel' and 'Save' buttons, with the 'Save' button also highlighted by a red arrow.

STATUS	DEVICE NAME	MACHINE TYPE
Pending	device3	7246
Ready	device1	7246
Ready	device2	7246

DETAILS

Name: device3

MACHINE TYPE: 7246, SERIAL NUMBER: SSSSSSS, GROUP: no group

ACTIONS

- Activate
- Manually Activate
- Update Key**
- Restore Key Backup
- Re-sync
- Unclaim

Cancel Save

Step



ThinkShield Web application – Update Key

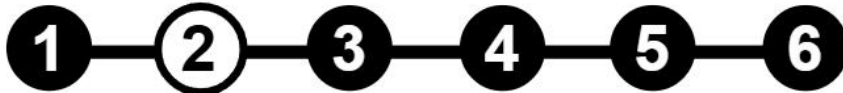
Enter the Activation Code from the defective system board and the Device key of the new system board, and then click **Update**.

- The Activation Code is printed on the asset tag and the flyer that was shipped with the server. **Asset tag and flyer**
- Users can scan the QR code on the new system board to get the Device key, or they can get the information using IPMI commands. For more information, refer to this Knowledge Base [link](#) or click the buttons below to see examples.

QR code

IPMI
command

Step



Secure Activation Code on the asset tag and flyer



A new field for the Secure Activation Code has been added to the pull-out asset tag and flyer.



Asset tag template showing fields for XCC, XCC MAC Address, IPv6, Link Local Address, and Secure Activation Code. The Secure Activation Code field is highlighted with a red box.

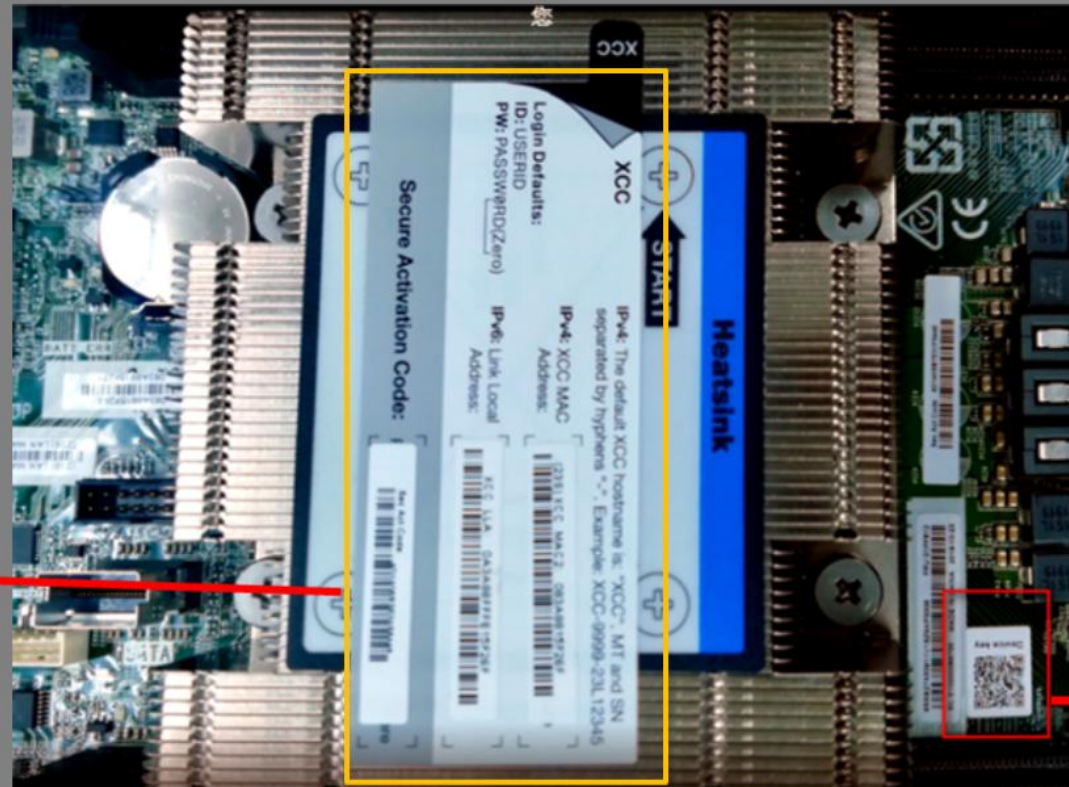


Lenovo ThinkShield flyer template. It includes a barcode, PN: SP47AS8937, and a note: "ATTENTION: DO NOT THROW AWAY". The flyer contains two columns of information: "SE350 with Security Pack" and "SE350 Standard (Security Pack disabled)". Below this is a section for the "Secure Activation Code" with a table for "Server information for activation" (Machine Type, Serial Number, Secure Activation Code). There is also a section for "Service technician only" with a table for "New secure activation code after system board replacement". The flyer concludes with a note about the Secure Activation Code, a link for mobile app downloads, and logos for Google Play, App Store, and various Chinese app stores.



A QR code has been added to the system board. When you replace a system board, scan the QR code to get the Device key.

A Secure Activation Code sticker will be attached to the heat sink on the new system board. When you replace the system board, peel off the sticker and attach it to the pull-out asset tag.



Secure Activation
Code sticker

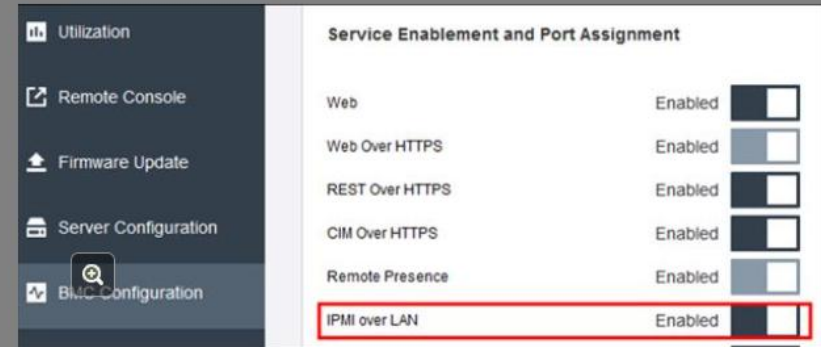


SE350 Device key
QR code

Using IPMI to get a Device key

Enable IPMI in XCC:

- Select **BMC Configuration** → **Network**, and then scroll down to the **Service Enablement and Port Assignment** section.
- Configure the state of the **IPMI over LAN** to **Enabled**.



Connect the laptop to the XCC port that has IPMITool installed, and then run the following command to get the Device key:

```
>ipmitool -I lanplus -U USERID -P PASSWORD -H 192.168.70.125 raw 0x3a 0xf2 0x6
```

Sample output:

```
54 41 8b d7 76 4f 1b 57 bf 63 0e 6b 31 ea 27 98
0b a4 29 e1 b2 12 86 c2 42 70 aa 2d 17 f7 26 ce
88 0c a8 a8 cb e9 0c cd 72 d2 74 05 f7 8b 7d a0
4f f9 3f 9e da 95 8f 1e a8 85 25 10 9a bd fe 9e
```

The SE350 Device key will be (64 bytes, 128 hex chars long):

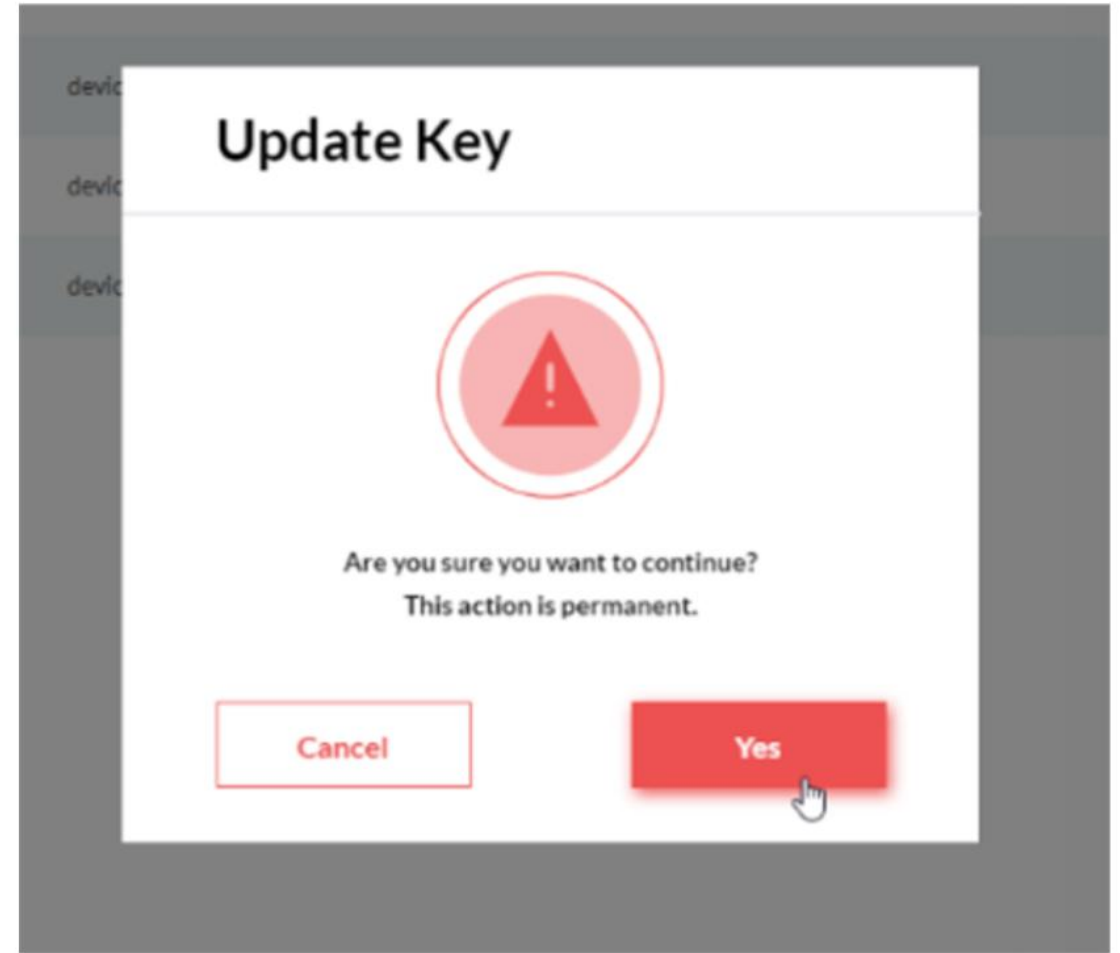
```
54418bd7764f1b57bf630e6b31ea27980ba429e1b21286c24270aa2d17f726ce880ca8a8cbe90ccd72d27405
f78b7da04ff93f9eda958f1ea88525109abdfef9e
```

Note: In this example, the XCC IP address is 192.168.70.125, the user ID is USERID, and the password is PASSWORD (zero, not capital o)

ThinkShield Web application – Update Key

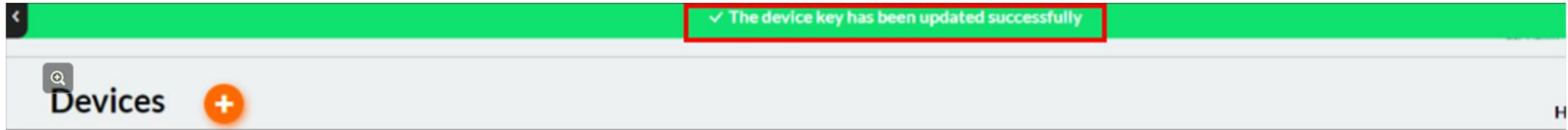
Confirm the action by clicking **Yes** in the **Update Key** pop-up window.

Step



ThinkShield Web application – Update Key

A message will be displayed stating that the Device key was successfully updated.



Note: Refer to the [ThinkShield Web application User Guide](#) for more information on how to configure each function in the ThinkShield Web application.

Step **1**—**2**—**3**—**4**—**5**—**6** 

ThinkShield Web application – Update Key

Check whether the device is displayed as Standard or Security Pack. If it should be Security Pack, go to the next step and use an IPMI command to update the device personality.

The screenshot displays the 'Devices' page in the ThinkShield Key Vault Portal. The left sidebar contains navigation links: Dashboard, Device Manager (selected), Apps, Users Manager, Organizations, and Activity History. The main content area shows a table of devices with columns: STATUS, MODE, DEVICE NAME, MACHINE TYPE, SERIAL NUMBER, and GROUP. The 'MODE' column is highlighted with a red box, indicating the current state of each device. The first two devices, 'NewDevice1' and 'NewDevice', are in 'Standard' mode, while the remaining five are in 'Security Pack' mode. The table also includes a 'Claim' column with checkboxes and a 'Group' column. The bottom of the page shows pagination information: '1 - 25 of 53 entries' and 'Show 25 entries'.

STATUS	MODE	DEVICE NAME	MACHINE TYPE	SERIAL NUMBER	GROUP
Unclaimed	Standard	NewDevice1	7Z46	PE02P301	
Unclaimed	Standard	NewDevice	7Z46	PE02LE01	
Unclaimed	Security Pack	Hsing3	7Z46	E1A1D03BCA	
Unclaimed	Security Pack	1033cfa9-bdaf-41aa-a80...	7Z46	USISO00001	
Unclaimed	Security Pack	eaf059f1-394e-4182-b9...	7D1X	COOLDEVICE11	
Unclaimed	Security Pack	1d2340ed-be3f-4295-88...	7Z46	BMC000009	
Unclaimed	Security Pack	f8b41d40-b05d-4801-81...	7D1X	KQ4CDVHGDFJ	
Unclaimed	Security Pack	83b69772-3370-4abc-8...	7Z46	ABASFNSDF	

Step



ThinkShield Web application – Update Key

Connect the laptop to the XCC port that has IPMITool installed, and then run the following command to set the device mode as Security Pack:

```
>ipmitool -I lanplus -U USERID -P PASSWORD -H 192.168.70.125 raw  
0x3a 0xf2 0x10 0xea 0x1
```

Note: In this example, the XCC IP address is 192.168.70.125, the user ID is USERID, and the password is PASSWORD (zero, not capital o).

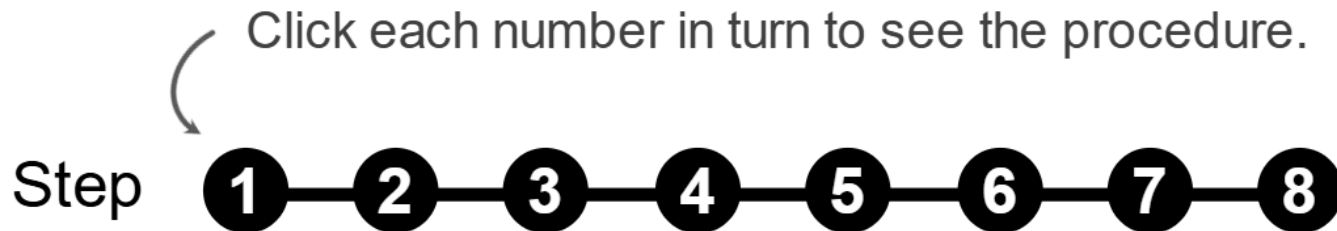
Note: Refer to the [ThinkShield Web application User Guide](#) for more information on how to configure each function in the ThinkShield Web application.

Step 

ThinkShield Web application – Transfer devices

The Transfer option allows users to send one or more devices to other organizations. The Transfer option includes the following three features:

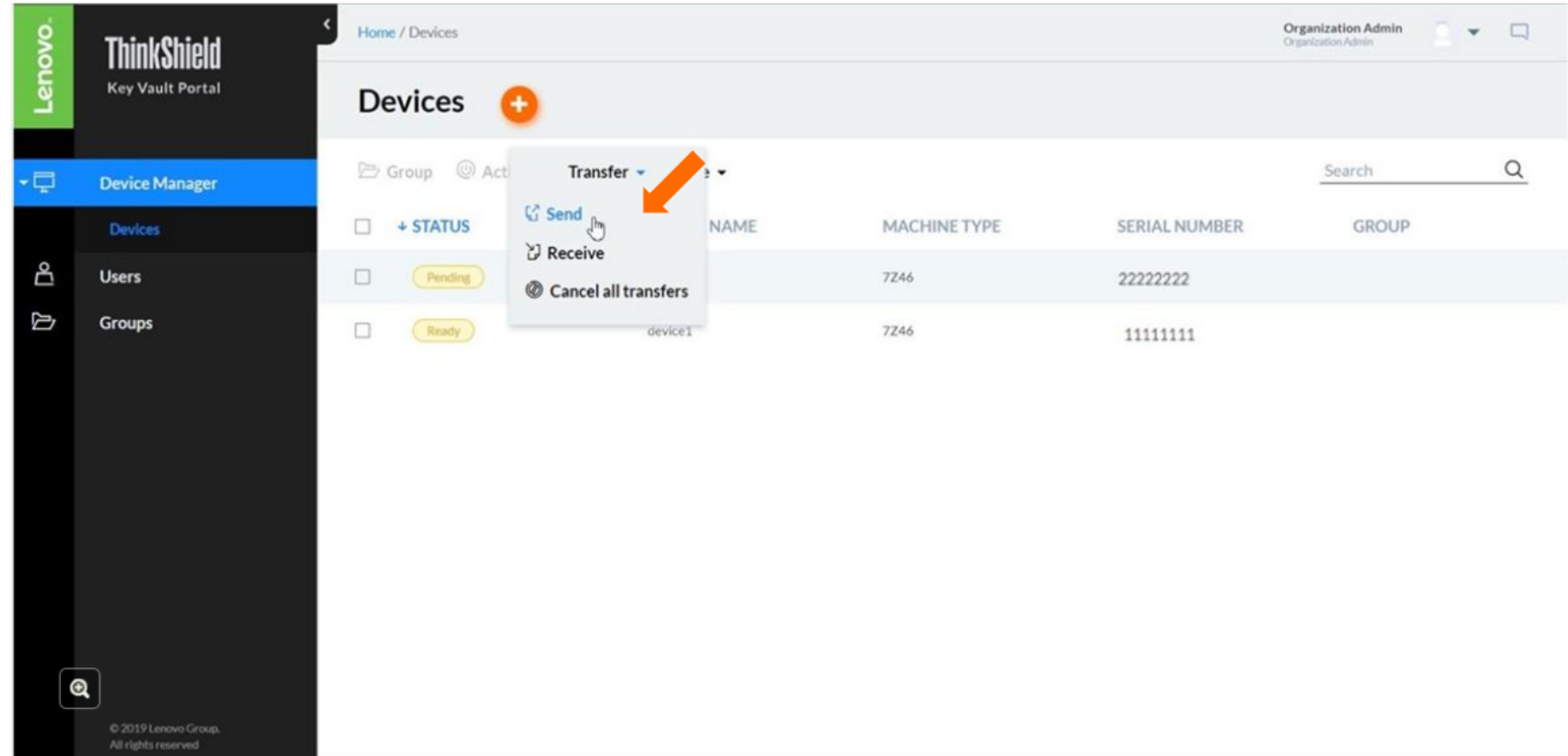
- Send
- Receive
- Cancel all transfers



ThinkShield Web application – Transfer devices

Send:

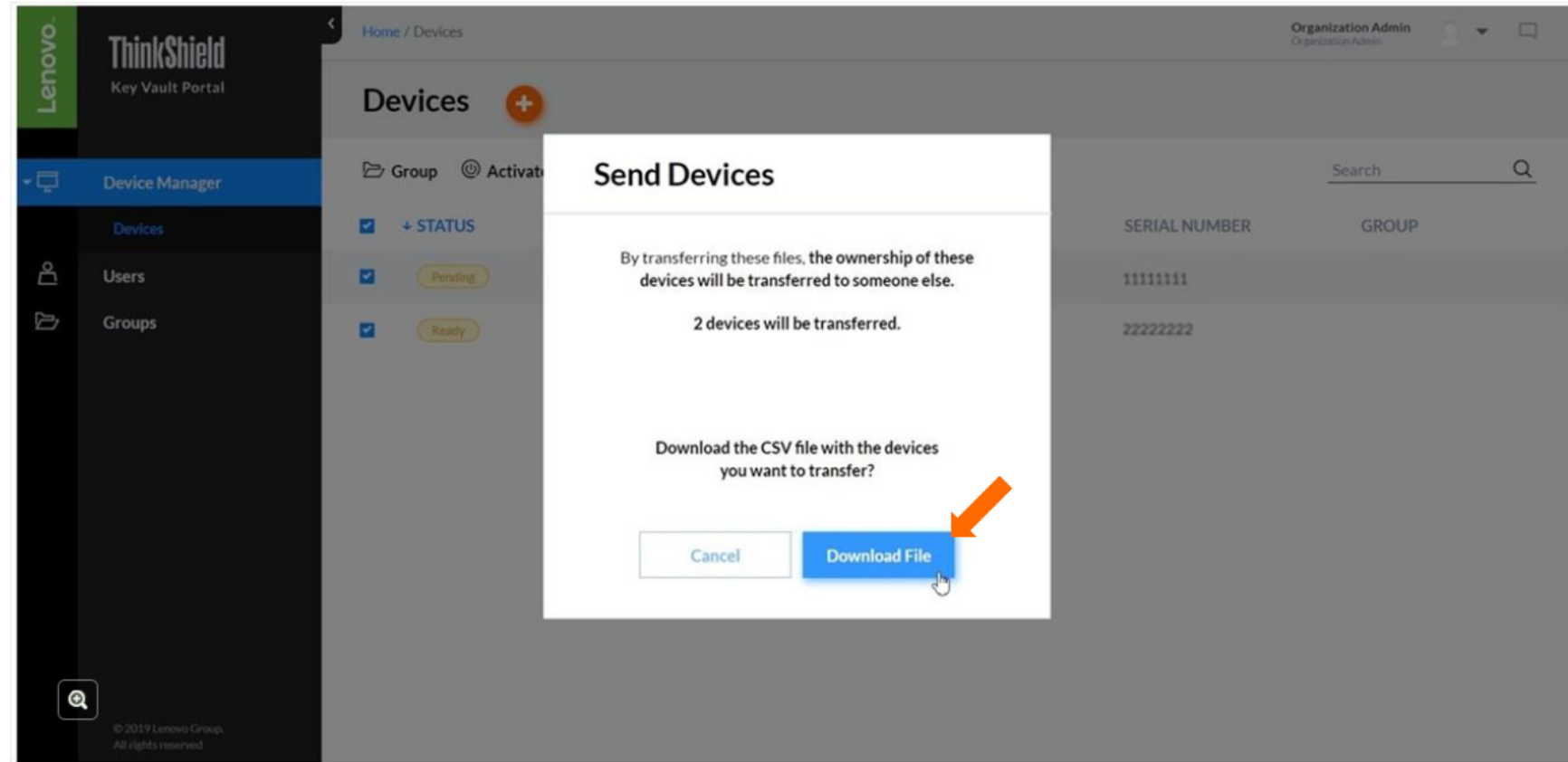
To send devices to another organization, select the check boxes next to the devices you want to transfer, and then select **Send** from the **Transfer** drop-down menu.



Step 1 — 2 — 3 — 4 — 5 — 6 — 7 — 8 

ThinkShield Web application – Transfer devices

When the **Send Devices** pop-up window is displayed, click **Download file**. A CSV file with the devices you want to transfer will be downloaded to your computer.



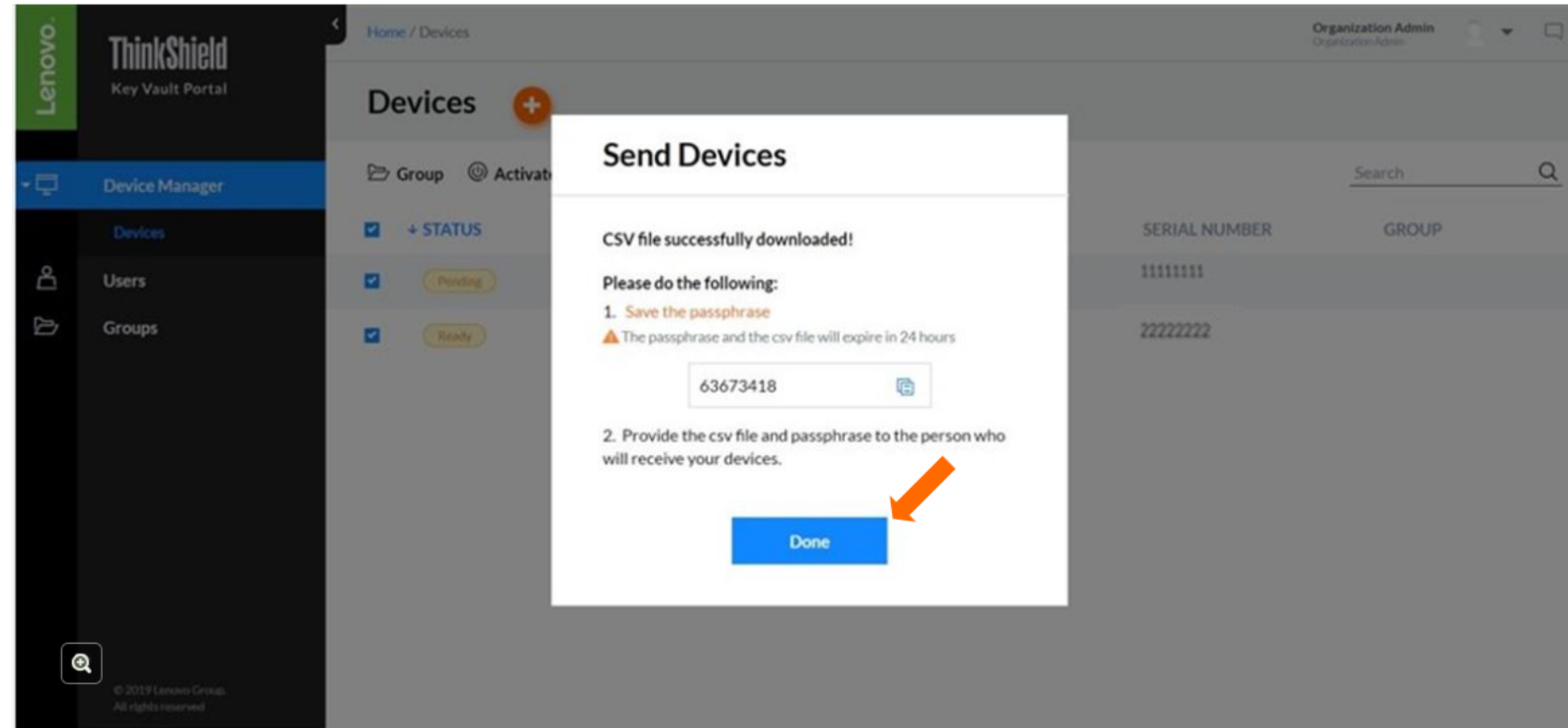
Step



ThinkShield Web application – Transfer devices

After the CSV file has been downloaded, the passphrase will be shown in the **Send Devices** pop-up window. Copy and save the passphrase, and then click **Done**. Send the downloaded CSV file and the passphrase to the person who will receive your devices.

Note: The passphrase and the CSV file will expire after 24 hours. Make sure you have sent them to the receiving person within 24 hours.



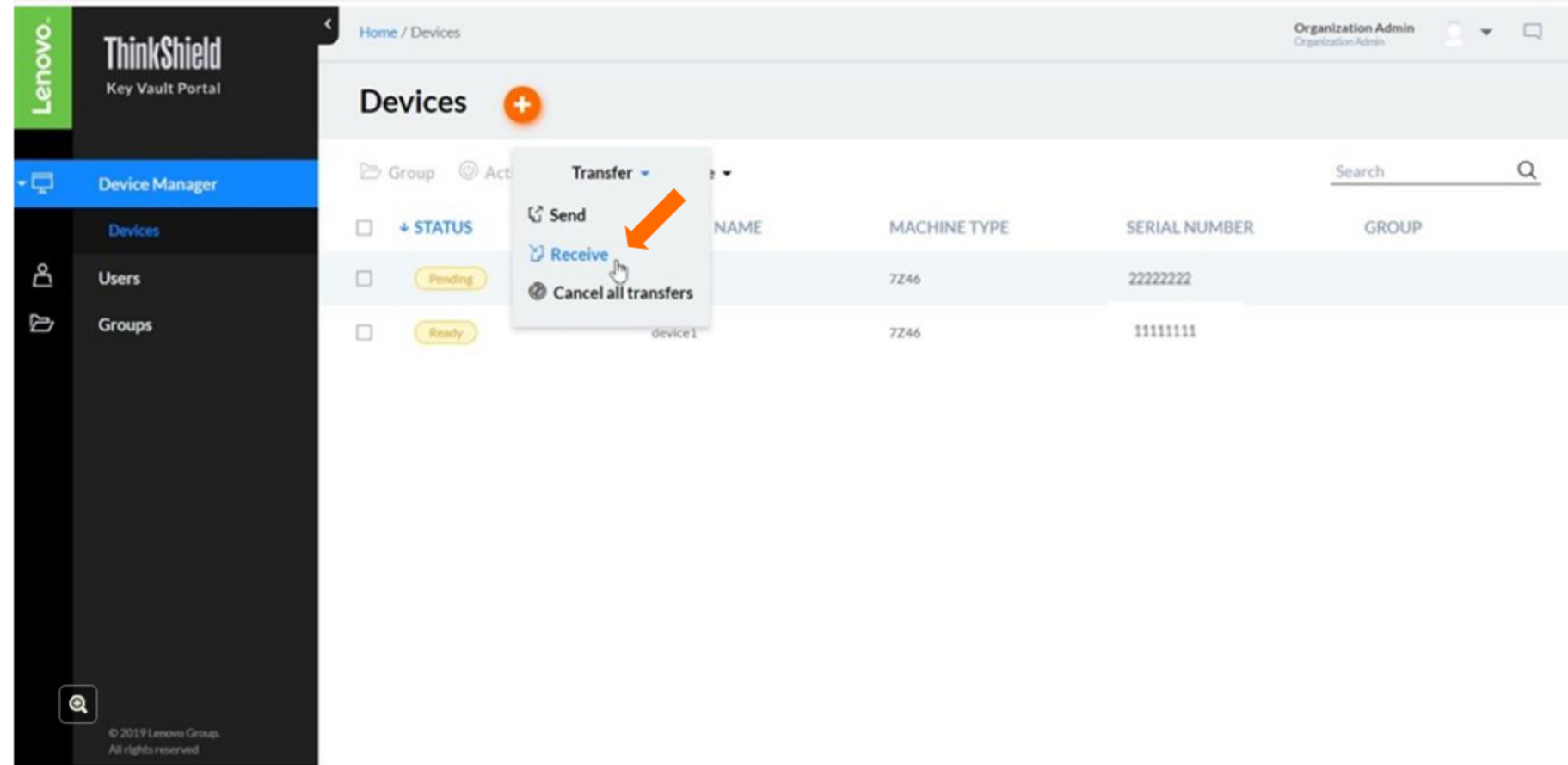
Step



ThinkShield Web application – Transfer devices

Receive:

To receive devices from another organization, select **Receive** from the **Transfer** drop-down menu.

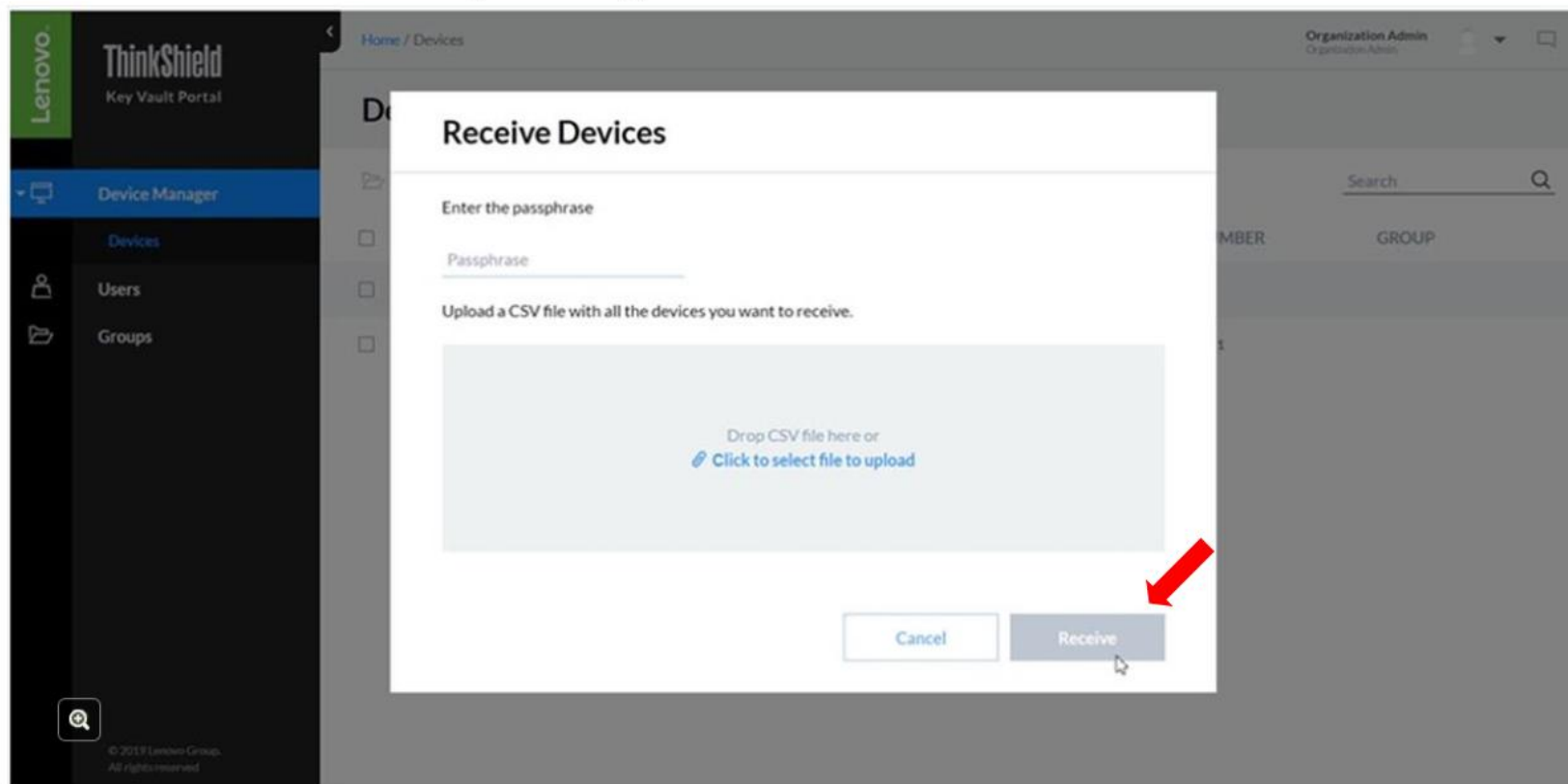


Step



ThinkShield Web application – Transfer devices

Enter the passphrase and upload the CSV file you received from the other organization. Click **Receive**, and the devices will be transferred to your organization.



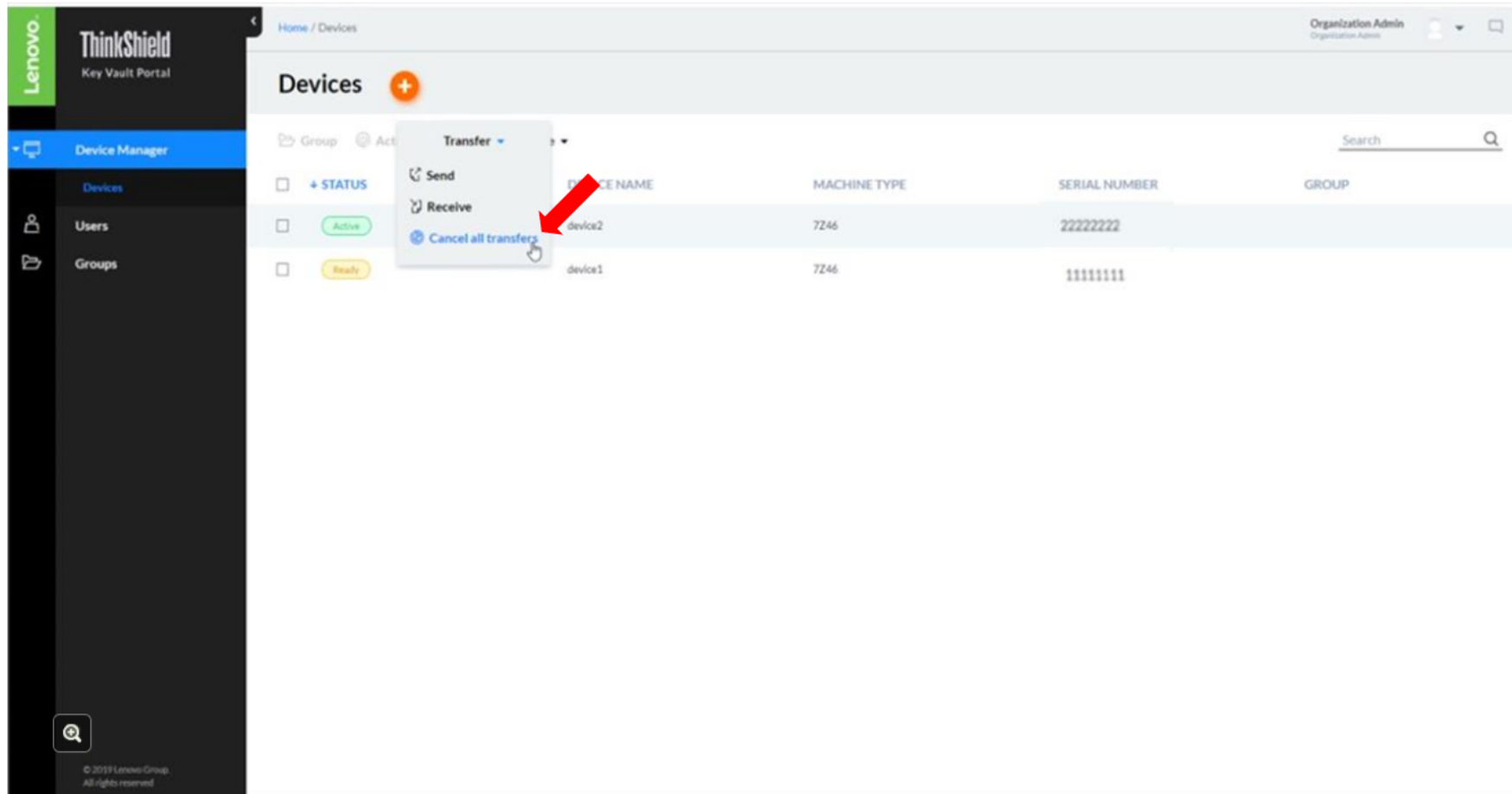
Step



ThinkShield Web application – Transfer devices

Cancel all transfers:

To cancel any transfers, select **Cancel all transfers** from the **Transfer** drop-down menu. This will stop all your current transfers and organizations will be unable to receive devices.

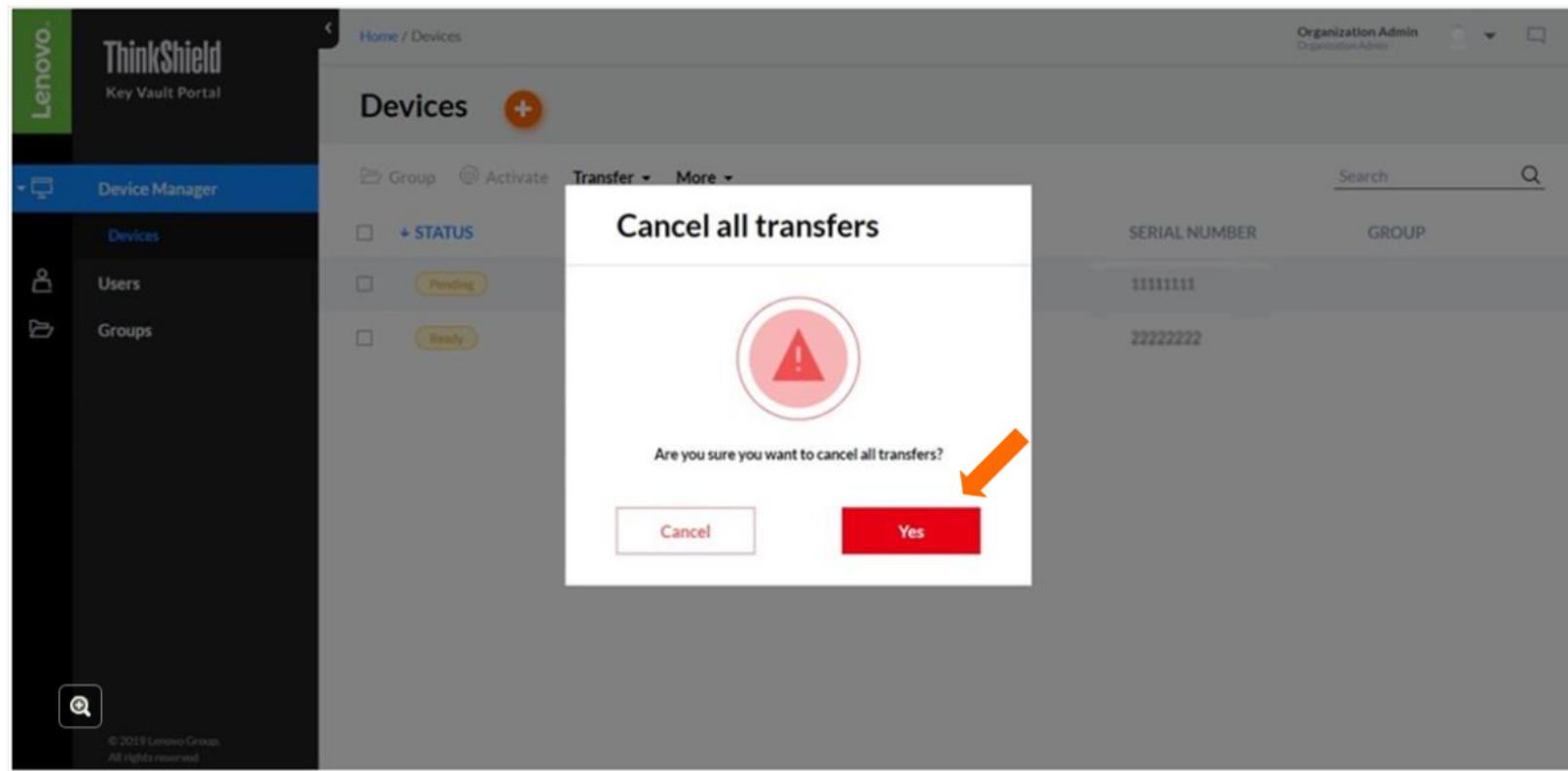


Step



ThinkShield Web application – Transfer devices

Confirm the action by clicking **Yes** in the **Cancel all transfers** pop-up window.

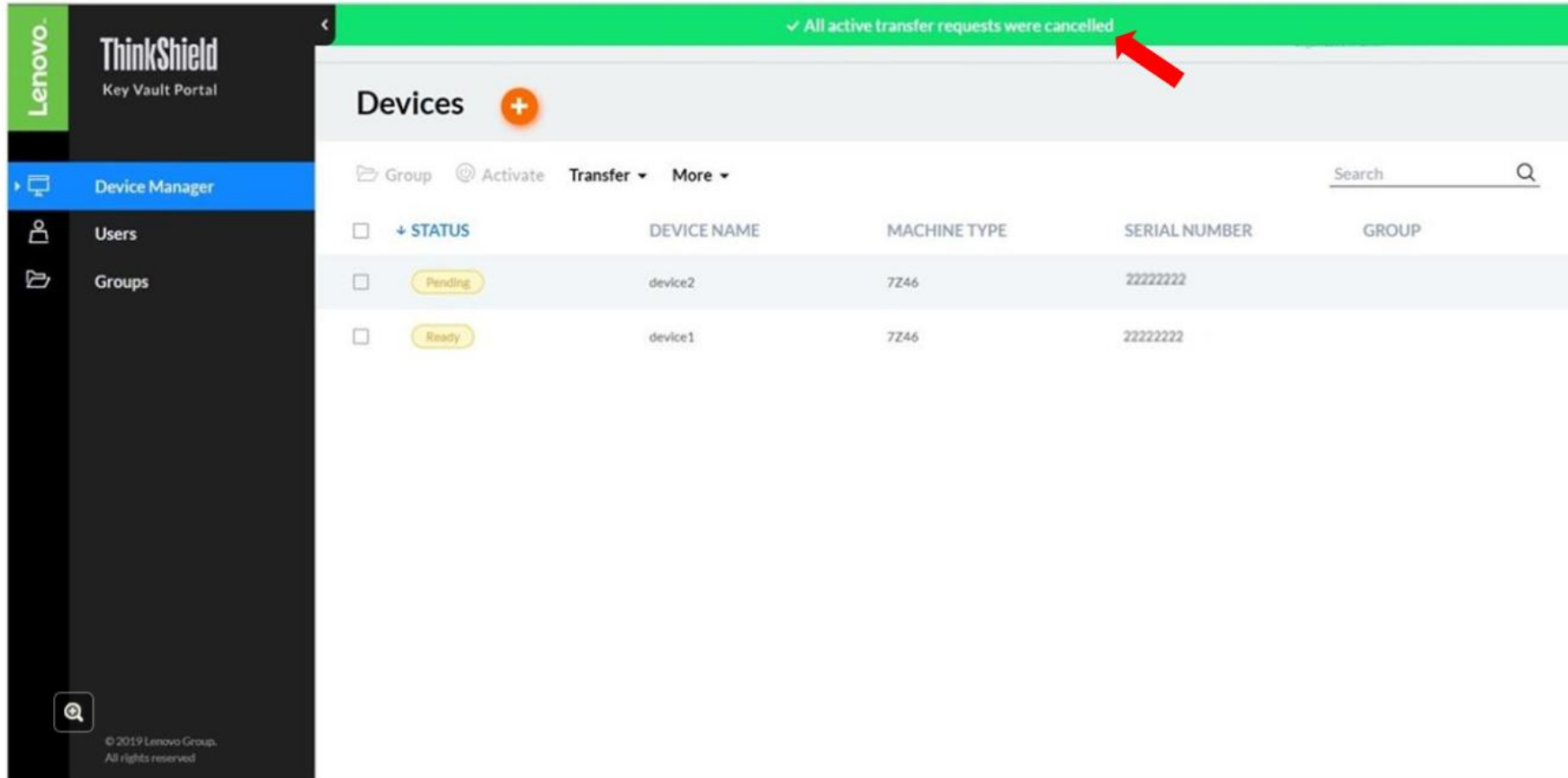


Step



ThinkShield Web application – Transfer devices

A message stating that **All active transfer requests were cancelled** will be displayed.



The screenshot shows the ThinkShield Key Vault Portal interface. A green banner at the top displays the message "✓ All active transfer requests were cancelled", with a red arrow pointing to it. The left sidebar contains navigation links for "Device Manager", "Users", and "Groups". The main content area is titled "Devices" and includes a table with columns for "STATUS", "DEVICE NAME", "MACHINE TYPE", "SERIAL NUMBER", and "GROUP". The table lists two devices: "device2" with status "Pending" and "device1" with status "Ready".

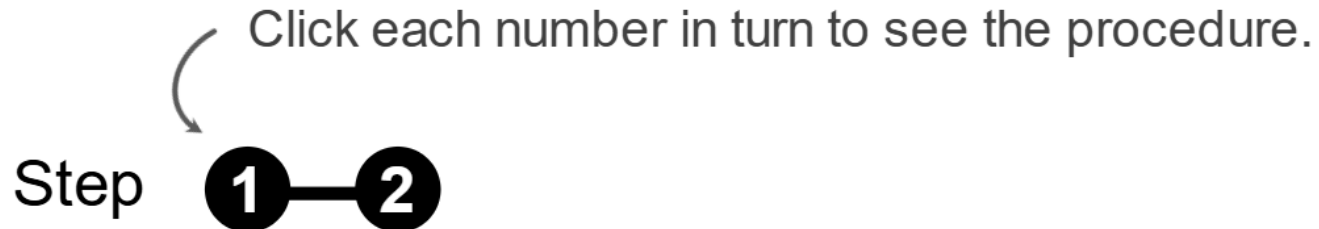
STATUS	DEVICE NAME	MACHINE TYPE	SERIAL NUMBER	GROUP
Pending	device2	7Z46	22222222	
Ready	device1	7Z46	22222222	

Step



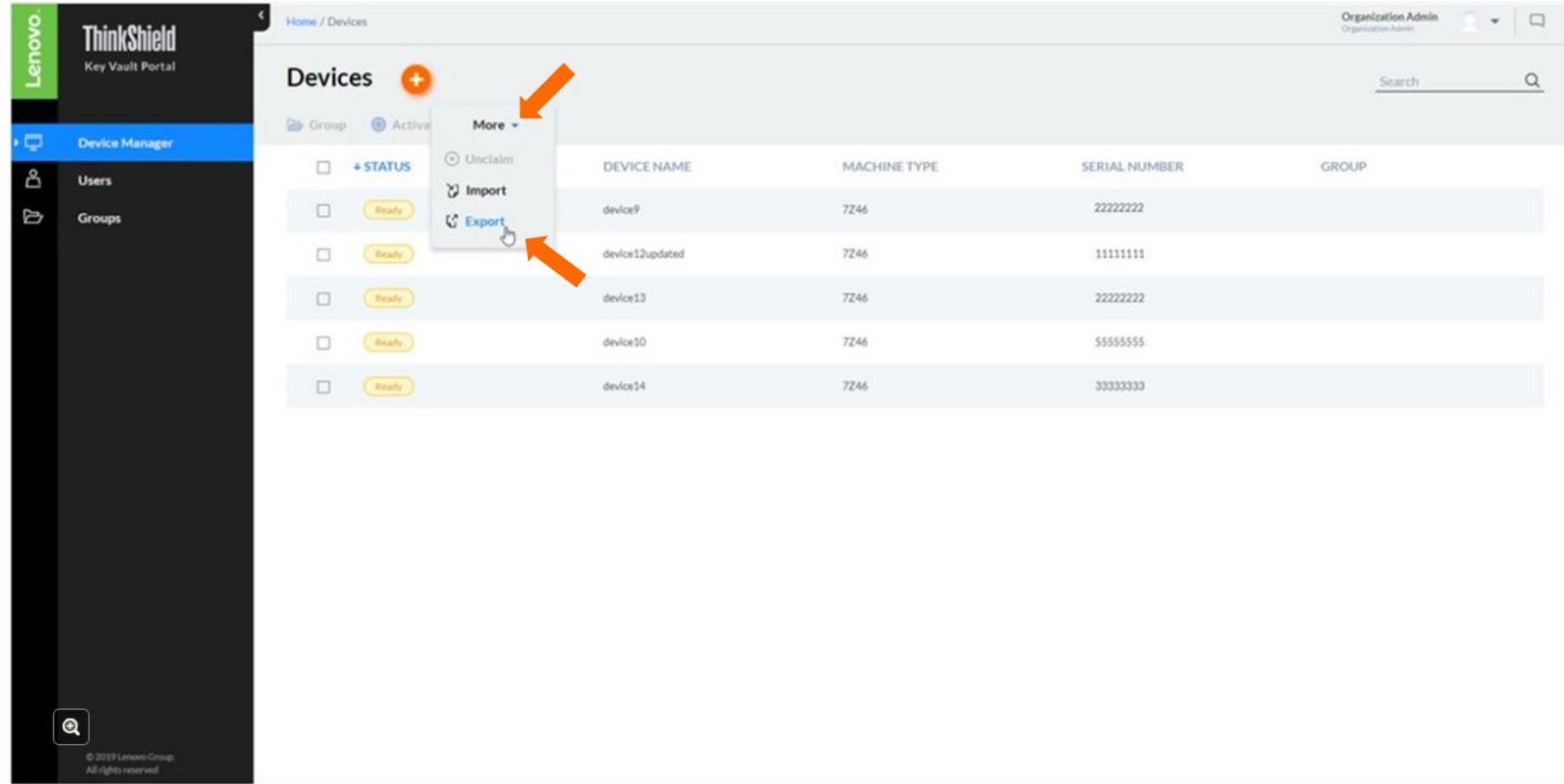
ThinkShield Web application – Export devices

The Export option allows users to download a file with a list of all their devices and their associated attributes.



ThinkShield Web application – Export devices

To export a list of devices, select **Export** from the **More** drop-down menu.



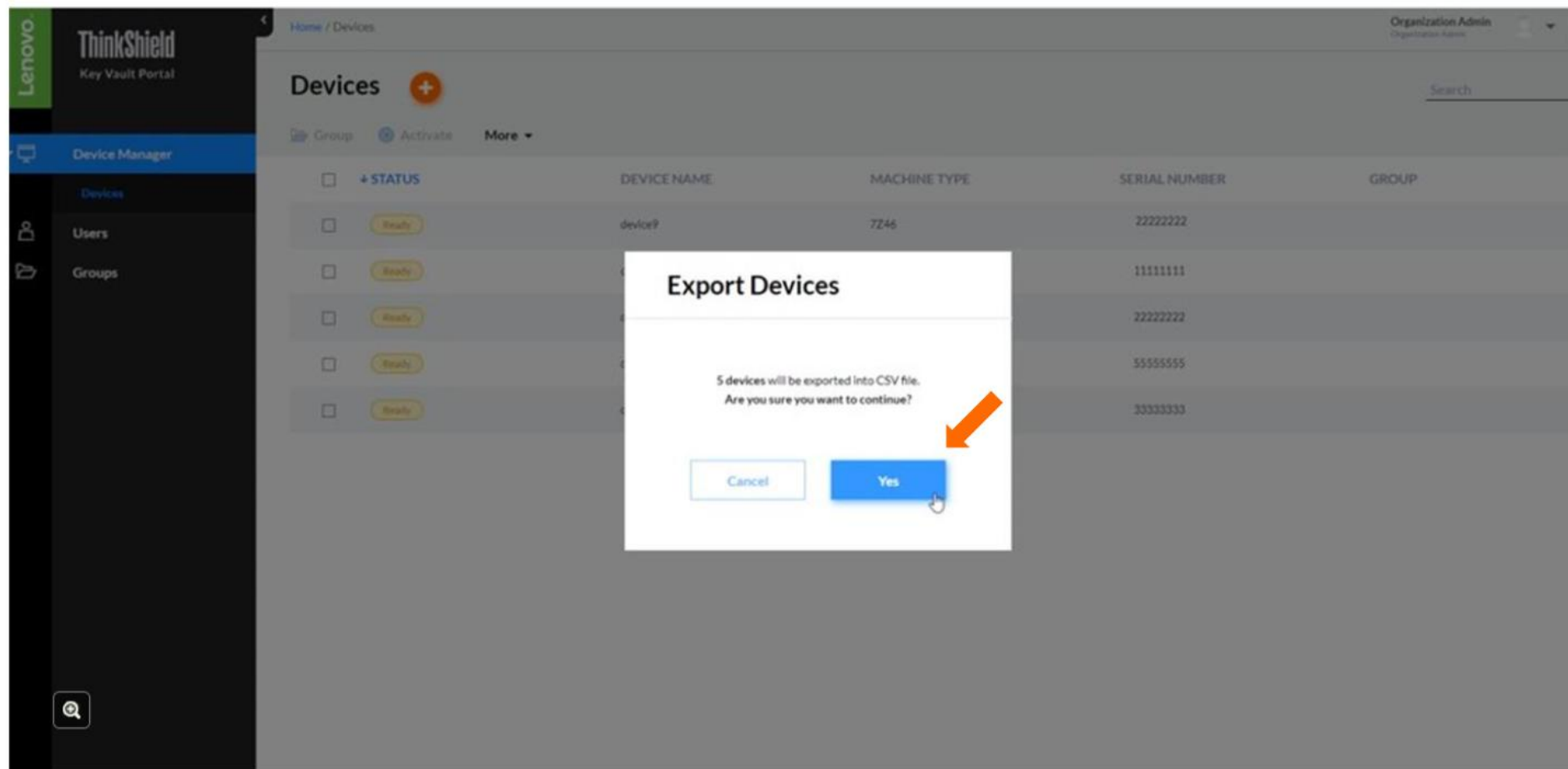
The screenshot shows the ThinkShield Key Vault Portal interface. On the left is a navigation sidebar with 'Device Manager' selected. The main area is titled 'Devices' and contains a table of device information. A 'More' dropdown menu is open over the table, with 'Export' highlighted by an orange arrow. Another orange arrow points to the 'More' button. The table has columns for Device Name, Machine Type, Serial Number, and Group. The first row shows 'device9' with Machine Type '7Z46' and Serial Number '22222222'. The second row shows 'device12updated' with Machine Type '7Z46' and Serial Number '11111111'. The third row shows 'device13' with Machine Type '7Z46' and Serial Number '22222222'. The fourth row shows 'device10' with Machine Type '7Z46' and Serial Number '55555555'. The fifth row shows 'device14' with Machine Type '7Z46' and Serial Number '33333333'. All devices have a status of 'Ready'.

Group	Active	Device Name	Machine Type	Serial Number	Group
☐	Ready	device9	7Z46	22222222	
☐	Ready	device12updated	7Z46	11111111	
☐	Ready	device13	7Z46	22222222	
☐	Ready	device10	7Z46	55555555	
☐	Ready	device14	7Z46	33333333	

Step 1 — 2 

ThinkShield Web application – Export devices

Click **Yes** in the **Export Devices** pop-up window to initiate the downloading process.



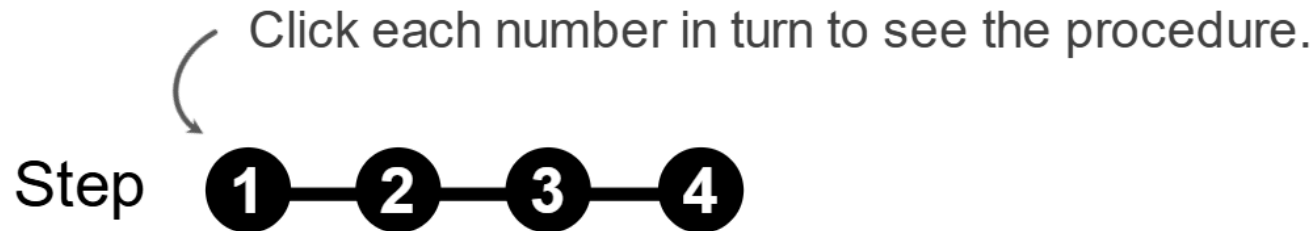
Step

1 — 2



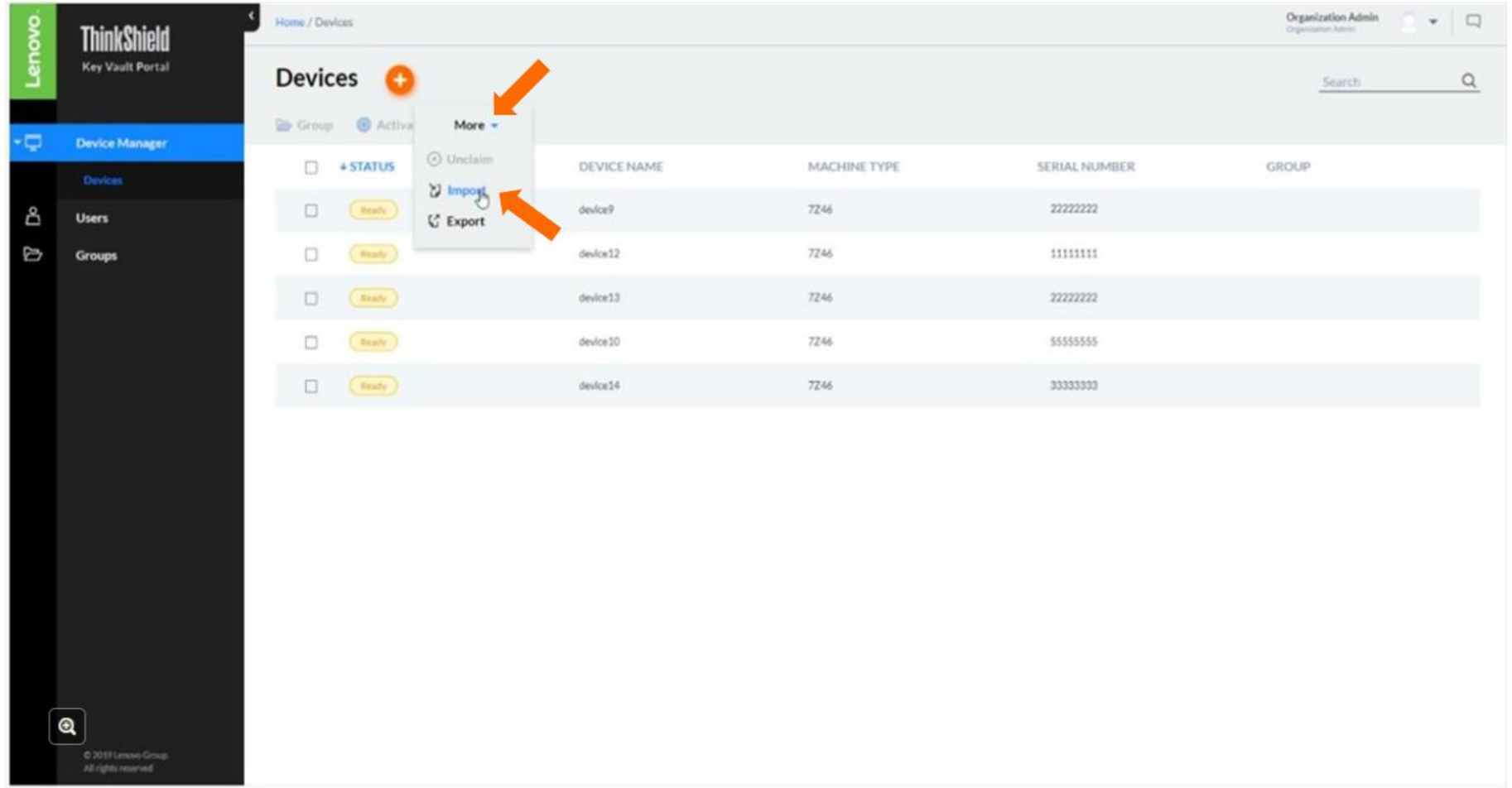
ThinkShield Web application – Import devices

The Import option allows users to upload updated device information.



ThinkShield Web application – Import devices

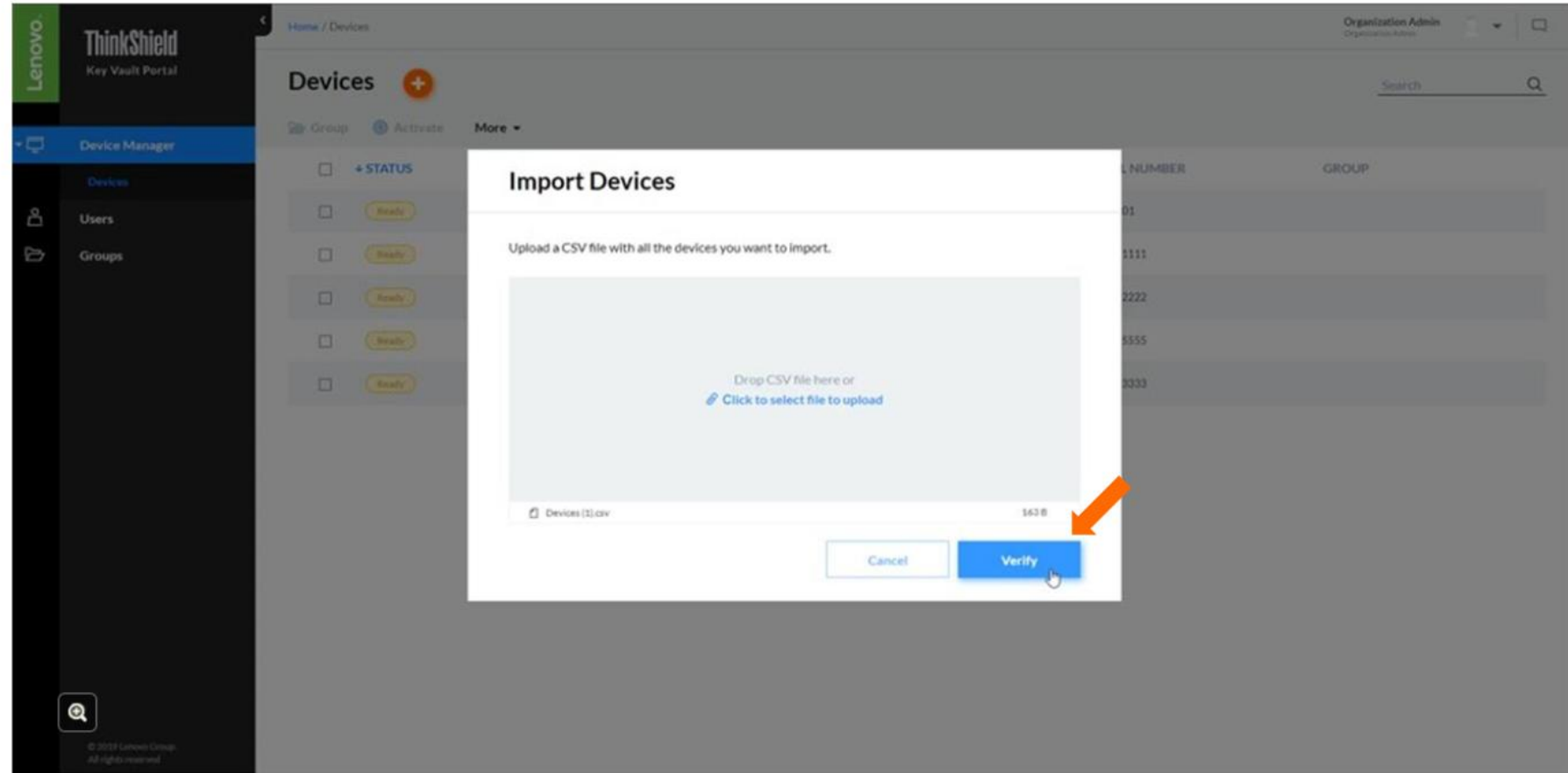
To start the importing process, select **Import** from the **More** drop-down menu.



Step ① — ② — ③ — ④ 

ThinkShield Web application – Import devices

Upload a CSV file with all the devices you want to update, and then click **Verify**.

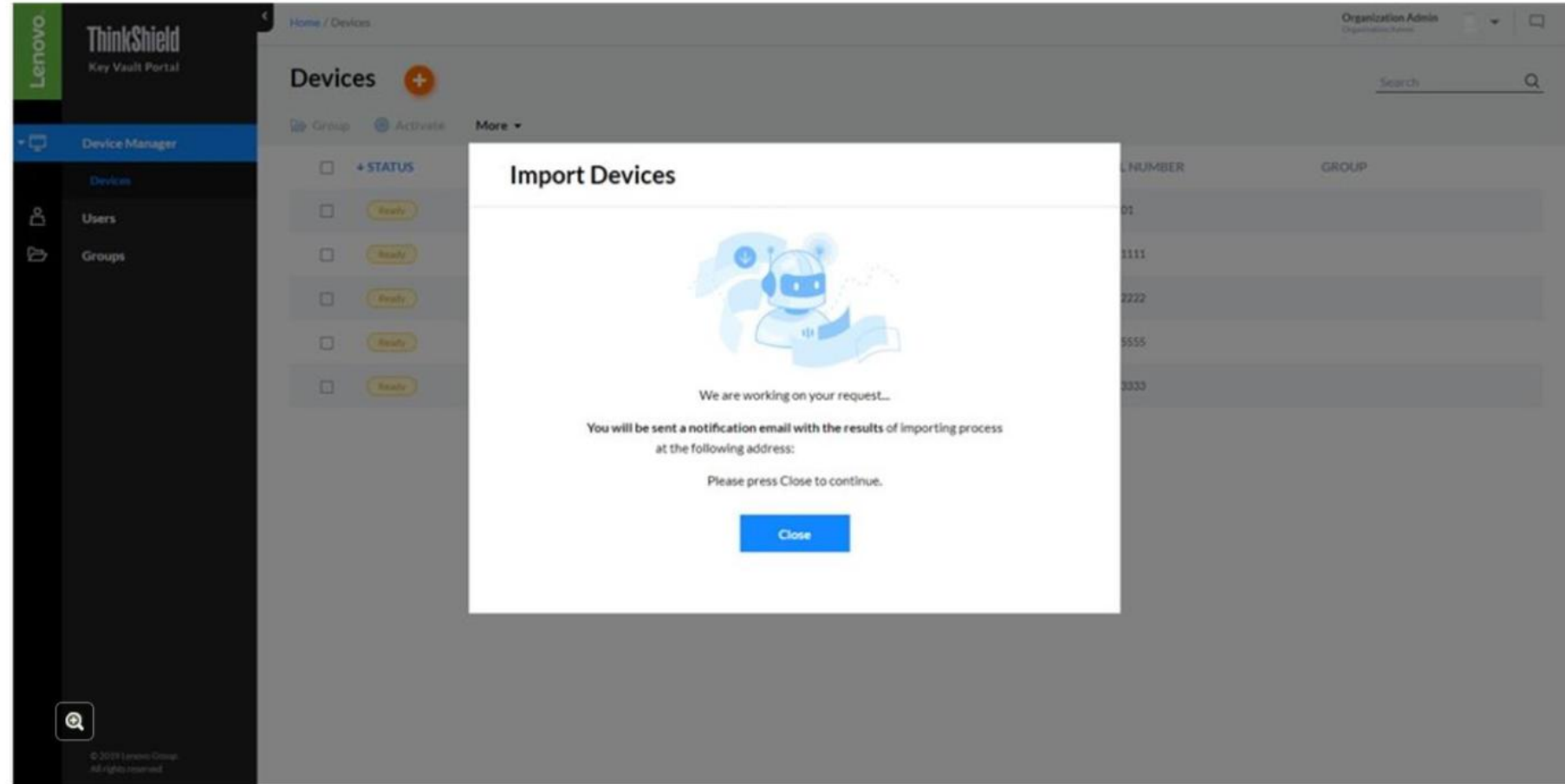


Step



ThinkShield Web application – Import devices

When the updating process is complete, a confirmation email will be sent to the email address associated with your profile.

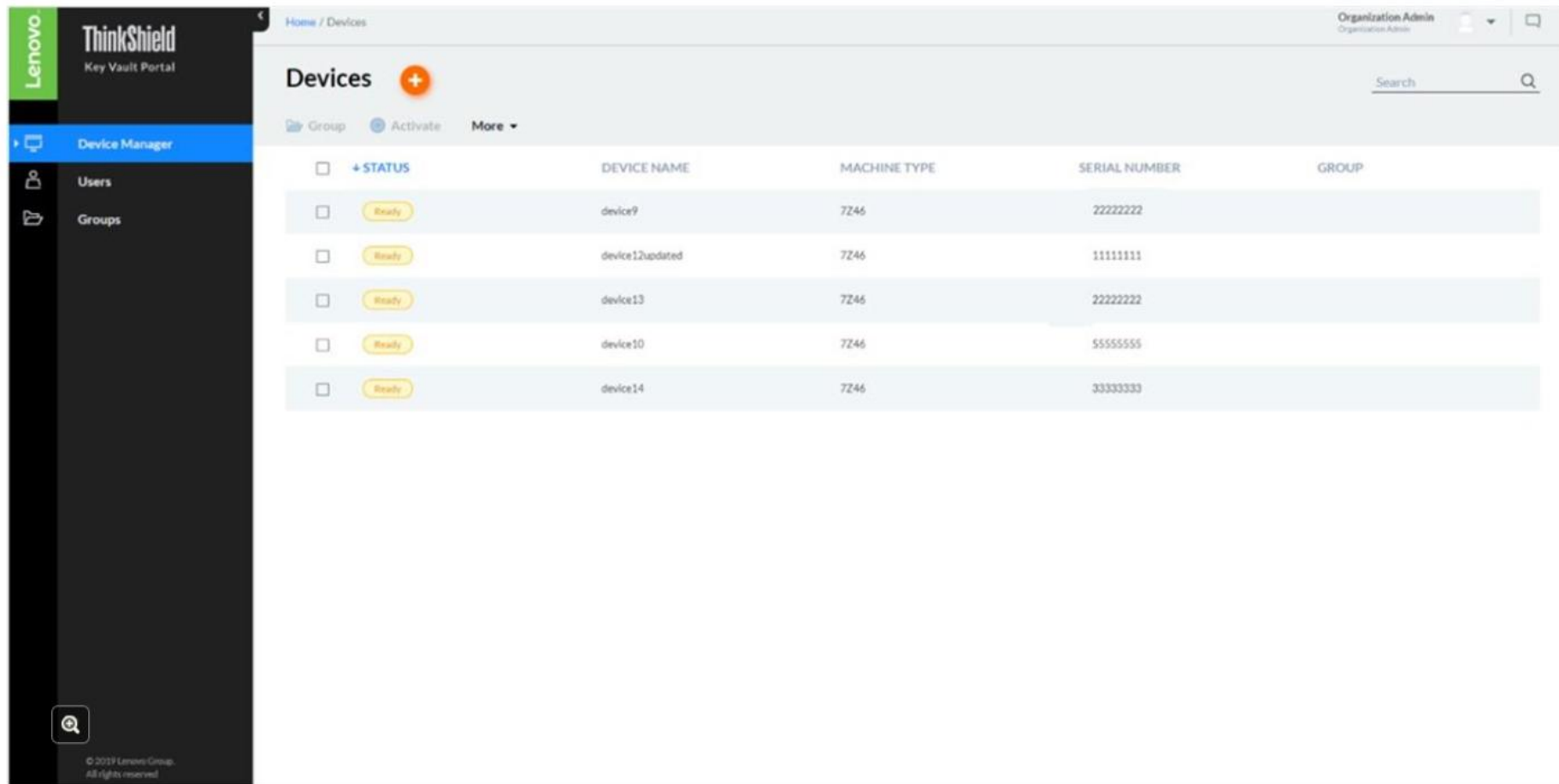


Step



ThinkShield Web application – Import devices

If the import is successful, the updates will be reflected in the **Devices** list.



The screenshot displays the ThinkShield Key Vault Portal interface. On the left is a dark sidebar with the Lenovo logo and navigation links for 'Device Manager', 'Users', and 'Groups'. The main content area is titled 'Home / Devices' and features a 'Devices' header with an orange plus icon. Below this is a table with columns for '+ STATUS', 'DEVICE NAME', 'MACHINE TYPE', 'SERIAL NUMBER', and 'GROUP'. The table contains five rows of device information, each with a 'Ready' status indicator. At the bottom of the sidebar, there is a copyright notice: '© 2019 Lenovo Group. All rights reserved.'

+ STATUS	DEVICE NAME	MACHINE TYPE	SERIAL NUMBER	GROUP
Ready	device9	7Z46	22222222	
Ready	device12updated	7Z46	11111111	
Ready	device13	7Z46	22222222	
Ready	device10	7Z46	55555555	
Ready	device14	7Z46	33333333	

Step



ThinkShield Web application – Export users

The Export option allows users to download a file with a list of all the users and their profile information.

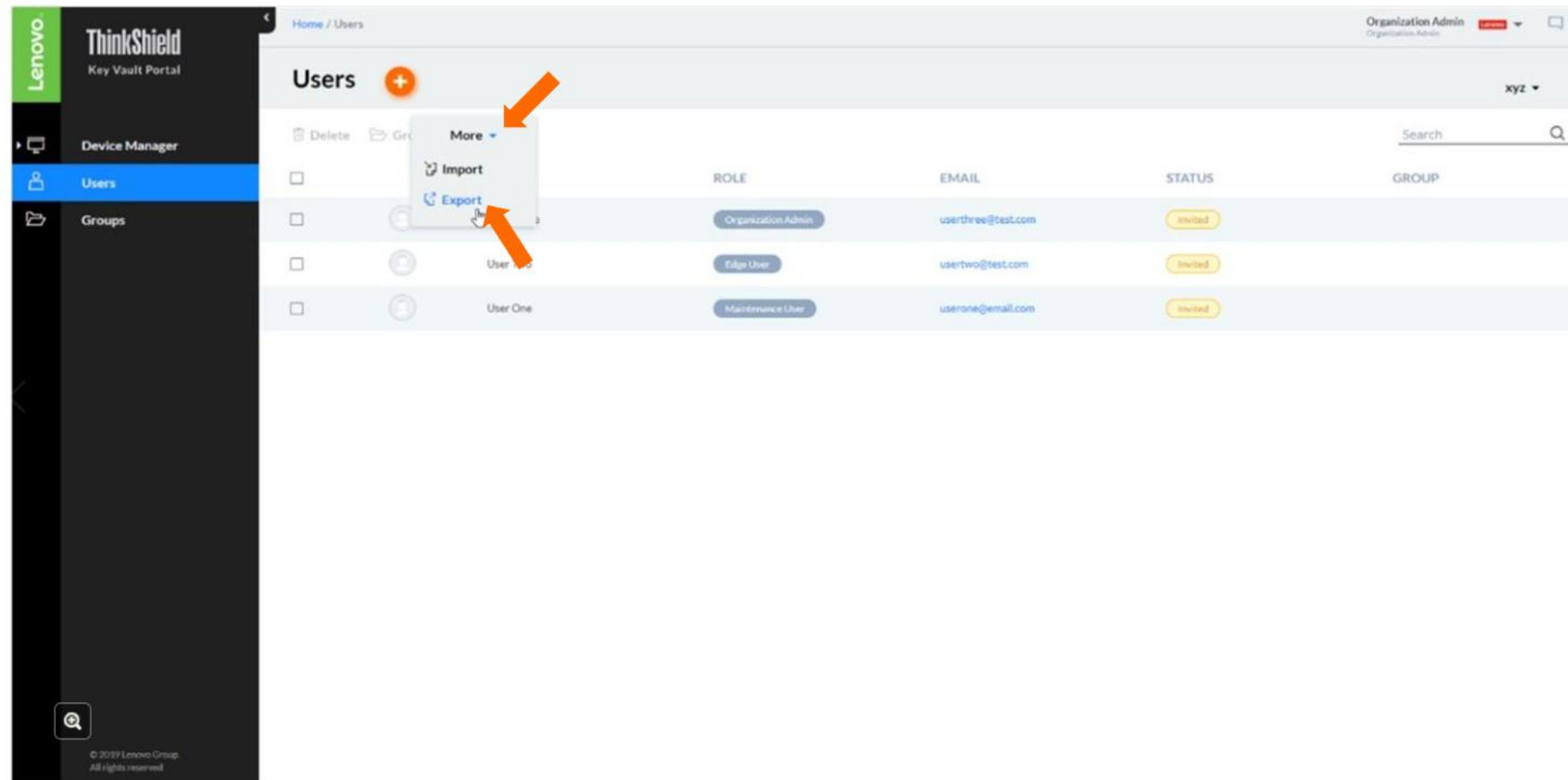
Click each number in turn to see the procedure.

Step

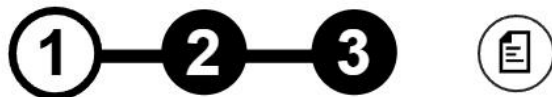


ThinkShield Web application – Export users

To export a list of users, select **Export** from the **More** drop-down menu.

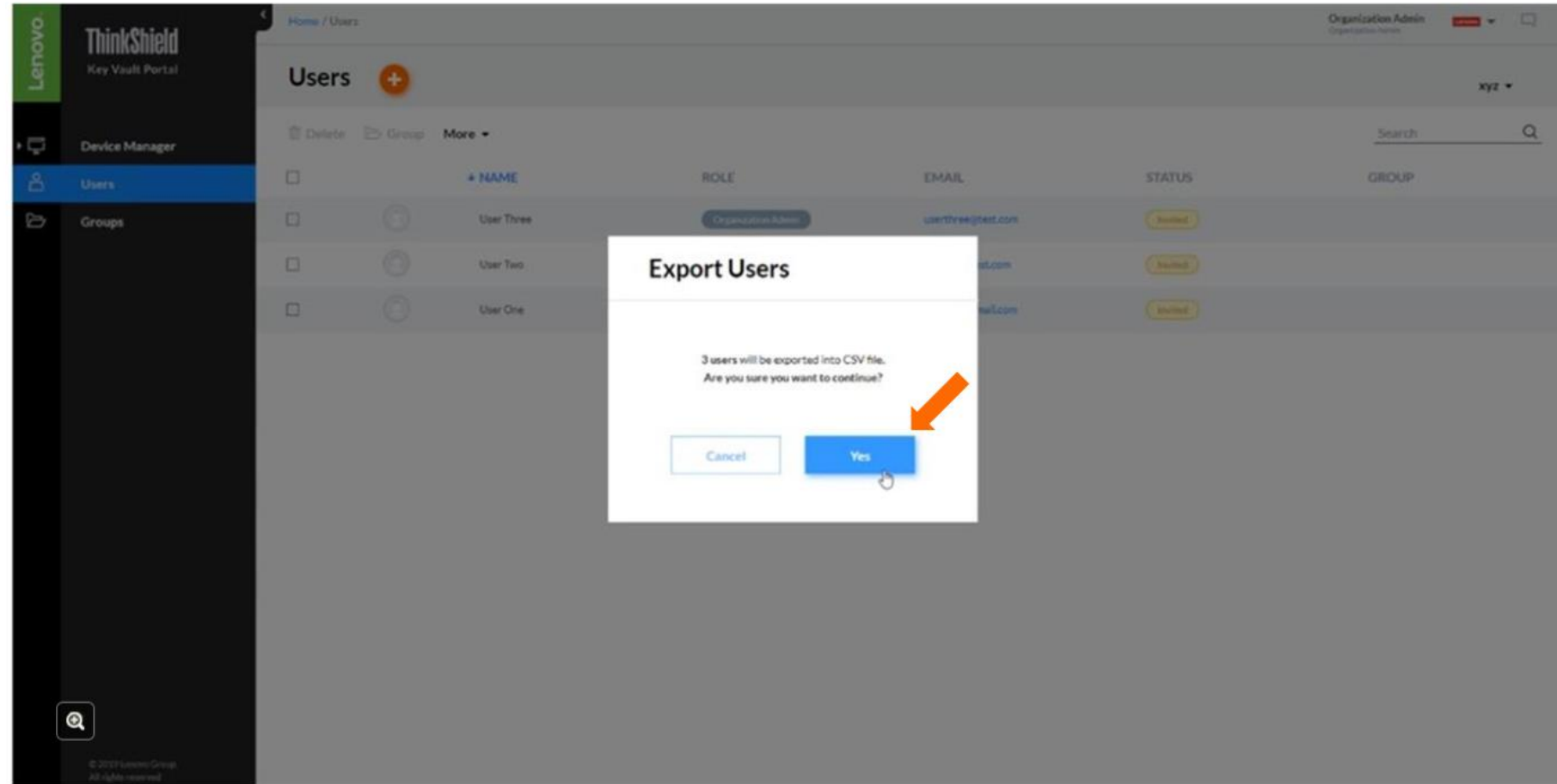


Step



ThinkShield Web application – Export users

Confirm the action by clicking **Yes** in the **Export Users** pop-up window.



Step **1** — **2** — **3** 

ThinkShield Web application – Export users

A CSV file with a list of the users will be downloaded to your computer.

Import

Character set: Cyrillic (Windows-1251) ▼

Language: Default - English (USA) ▼

From row: 1 ▼

Separator Options

☐ Fixed width ☒ Separated by

☒ Tab ☒ Comma ☒ Semicolon ☐ Space ☐ Other

☐ Merge delimiters ☐ Trim spaces String delimiter: " ▼

Other Options

☐ Format quoted field as text ☐ Detect special numbers

Fields

Column type: ▼

	Standard	Standard	Standard	Standard	Standard
1	First Name	Last Name	Role	Email	Action
2	User	One	Edge User	userone@test.com	
3	User	Three	Maintenance User	userthree@test.com	
4	User	Two	Organization Admin	usertwo@test.com	

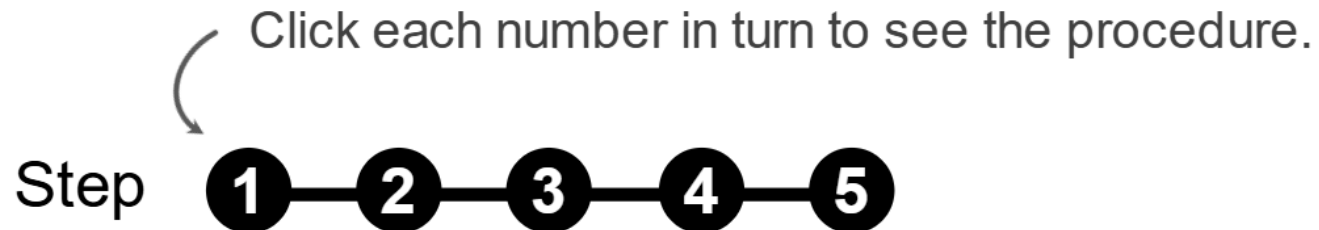
Help OK Cancel

Step



ThinkShield Web application – Import users

The Import option allows users to upload updated user information.



ThinkShield Web application – Import users

To edit a user list:

- Open the CSV file downloaded during the Export users procedure.
- Make any necessary changes to the list, and then save the file.
 - **Note:** The two available actions are updating a user's information or deleting a user from the list.
- Put **update** in the **Action** field next to any users whose information you want to update. Editable fields are **First Name**, **Last Name**, and **Role**.
- Put **delete** in the **Action** field next to any users you want to remove from the organization.

	A	B	C	D	E
1	First Name	Last Name	Role	Email	Action (update/delete)
2	User	Three	Edge User	<u>userthree@test.com</u>	update
3	User	Two	Edge User	<u>usertwo@test.com</u>	
4	User	One	Maintenance User	<u>userone@email.com</u>	delete
5					
6					
7					
8					
9					

Step

1

2

3

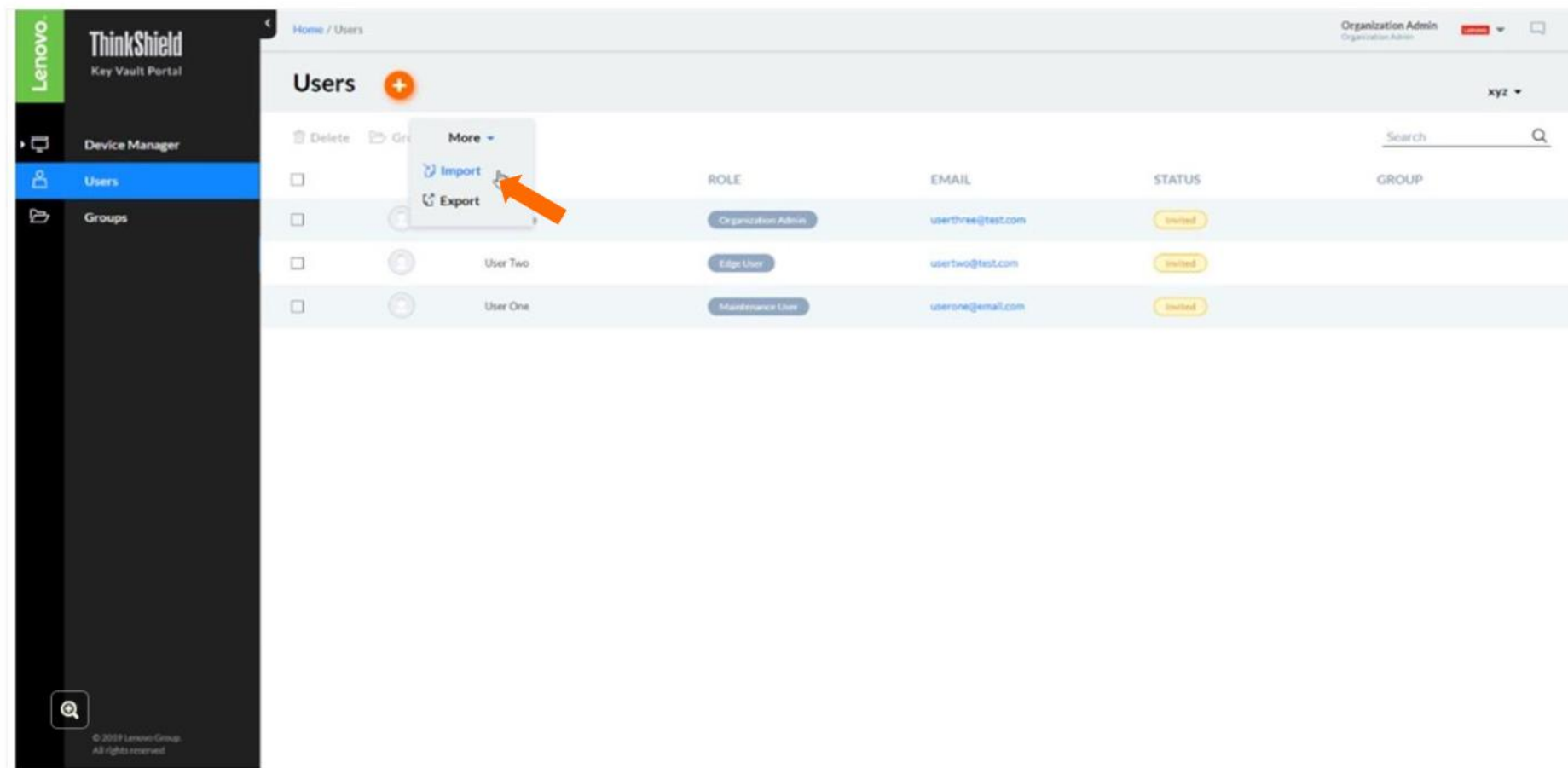
4

5



ThinkShield Web application – Import users

To start the importing process, select **Import** from the **More** drop-down menu.

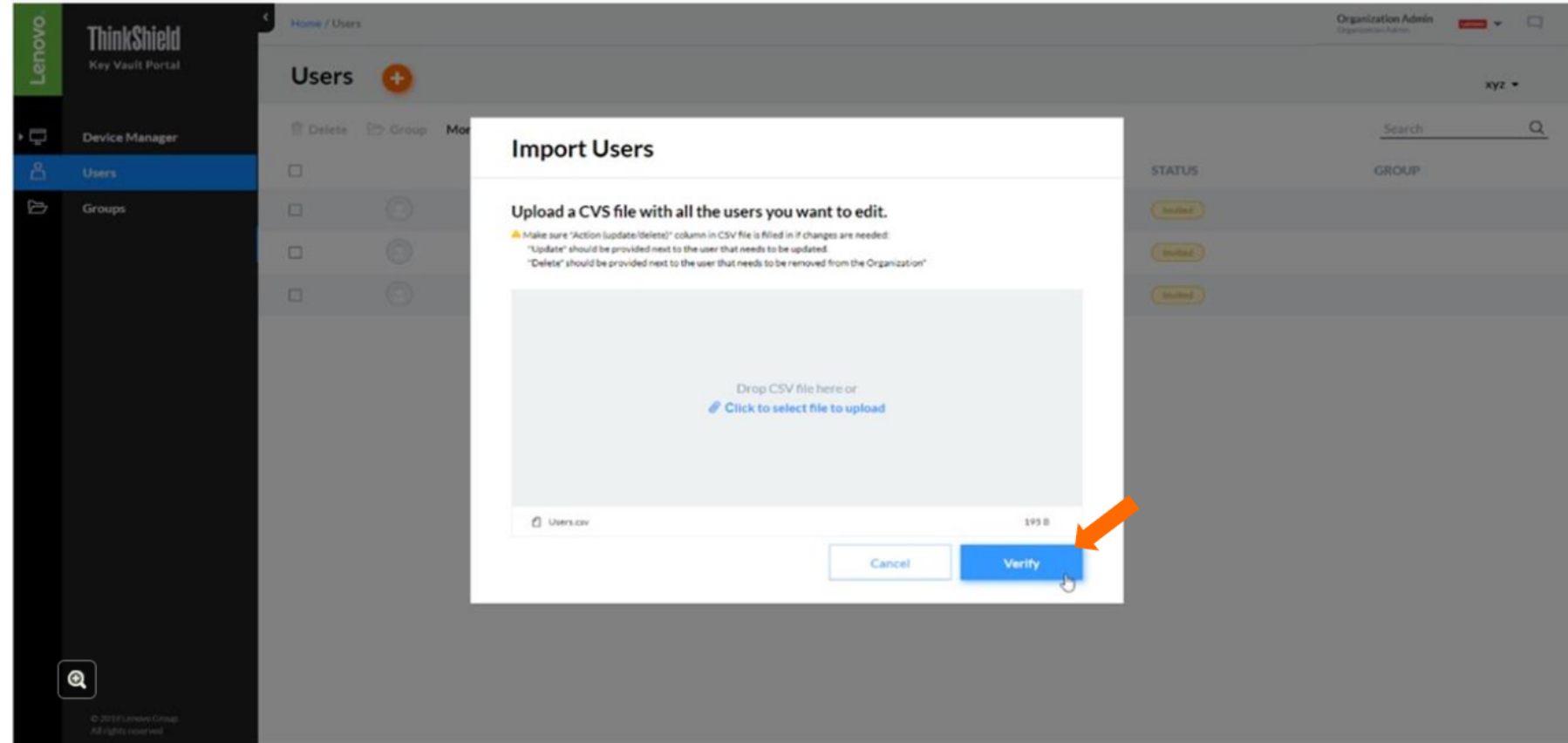


Step



ThinkShield Web application – Import users

Upload a CSV file to the Portal. Click **Verify** after the file has been uploaded.

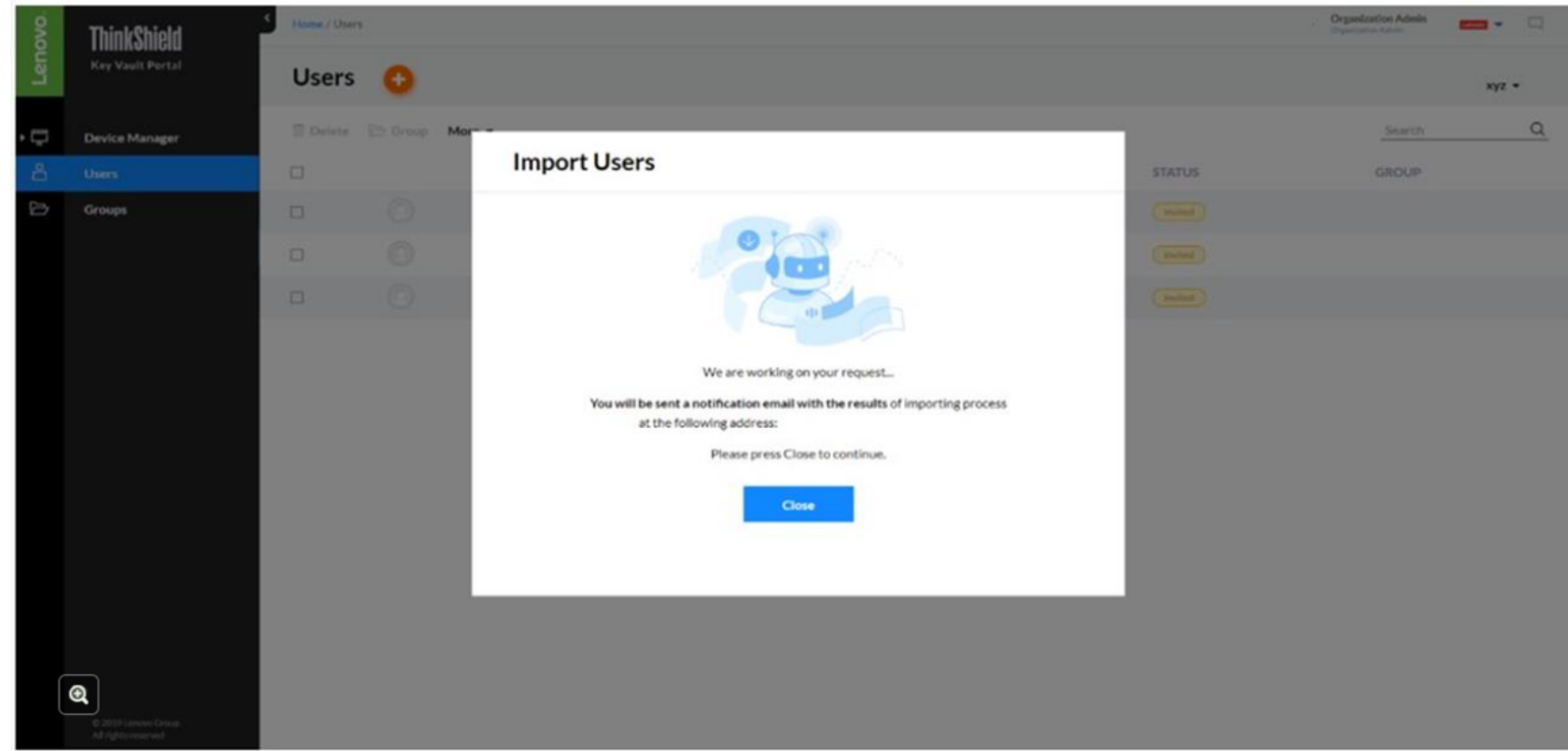


Step

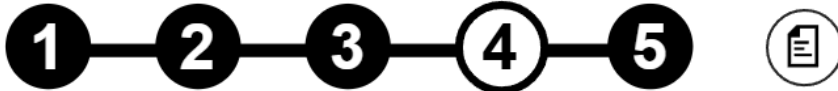


ThinkShield Web application – Import users

When the verification process is over, a confirmation email will be sent to the email address associated with your profile.

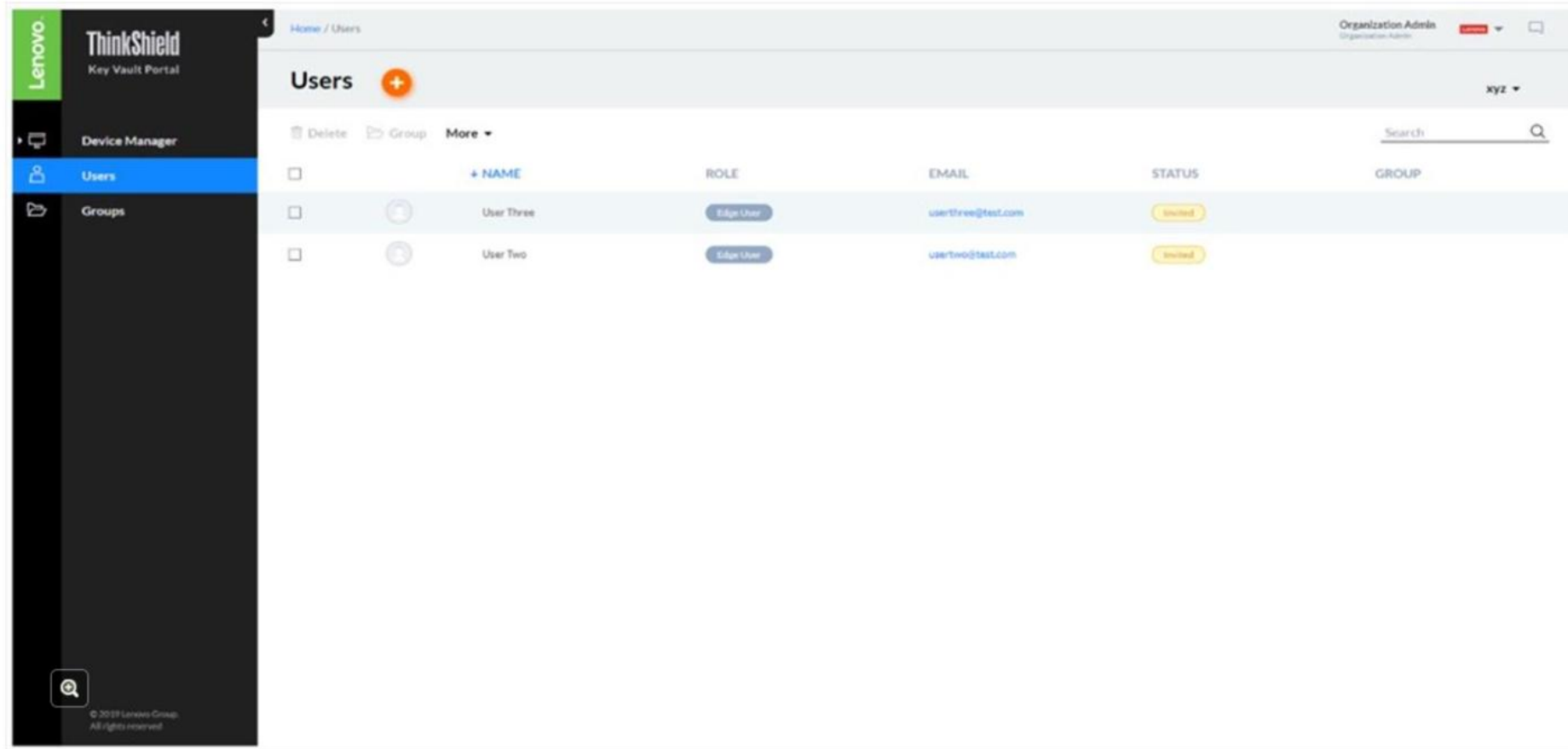


Step



ThinkShield Web application – Import users

The **Users** list will be updated.

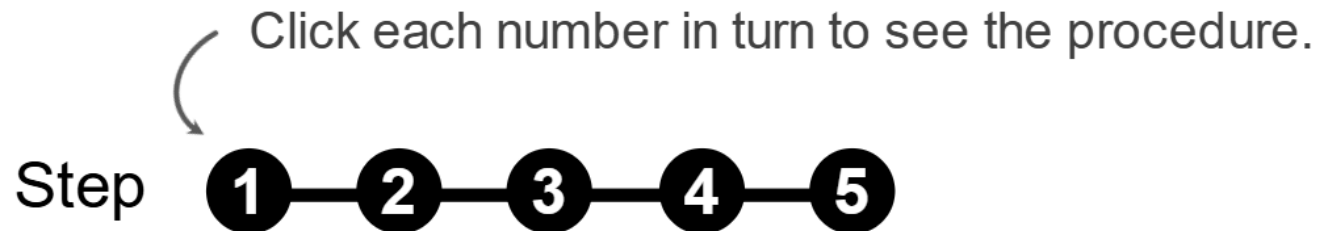


Step



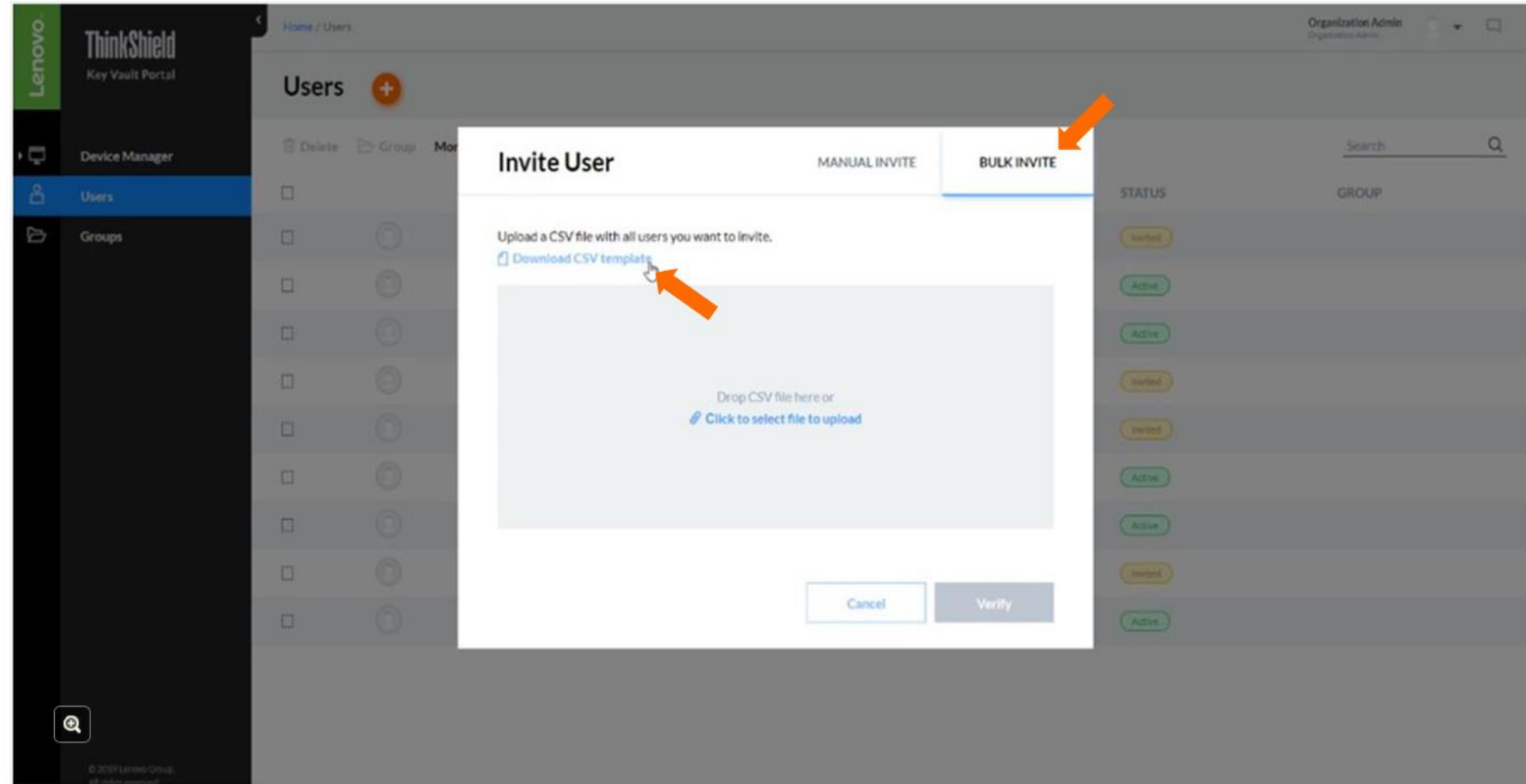
ThinkShield Web application – Bulk Invite users

The Bulk Invite option allows users to invite up to 1000 users at a time.

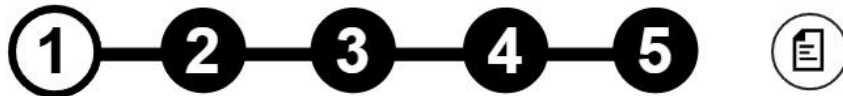


ThinkShield Web application – Bulk Invite users

Select the **BULK INVITE** tab on the **Invite User** pop-up window, and then download the CSV template.



Step

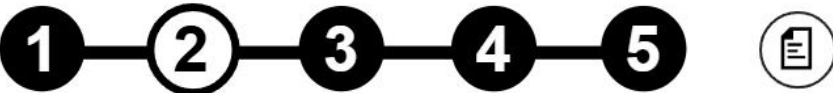


ThinkShield Web application – Bulk Invite users

Fill in the **First Name**, **Last Name**, **Role**, and **Email** fields.

	A	B	C	D
1	First Name	Last Name	Role	Email
2	User	One	Edge User	userone@test.com
3	User	Two	Organization Admin	usertwo@test.com
4	User	Three	Maintenance User	userthree@test.com
5				
6				
7				
8				
9				
10				

Step



Import

Character set: **Cyrillic (Windows-1251)**

Language: **Default - English (USA)**

From row: **1**

Separator Options

☐ Fixed width ☒ Separated by

☒ Tab ☒ Comma ☒ Semicolon ☐ Space ☐ Other

☐ Merge delimiters ☐ Trim spaces String delimiter: **"**

Other Options

☐ Format quoted field as text ☐ Detect special numbers

Fields

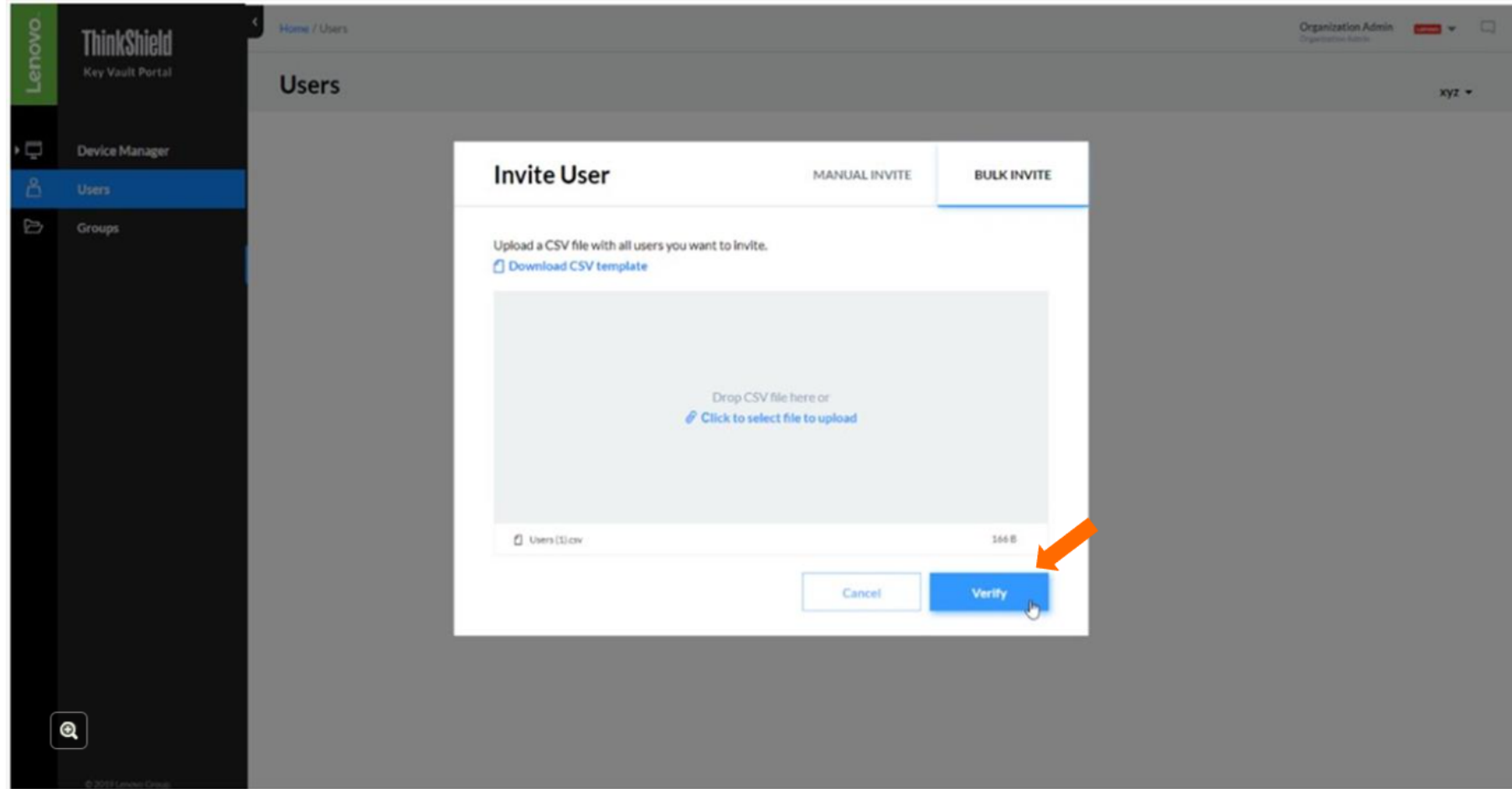
Column type: **Standard**

	Standard	Standard	Standard	Standard
1	First Name	Last Name	Role	Email

Help **OK** **Cancel**

ThinkShield Web application – Bulk Invite users

Upload the CSV file to the Portal. Click **Verify** after the file has been uploaded.

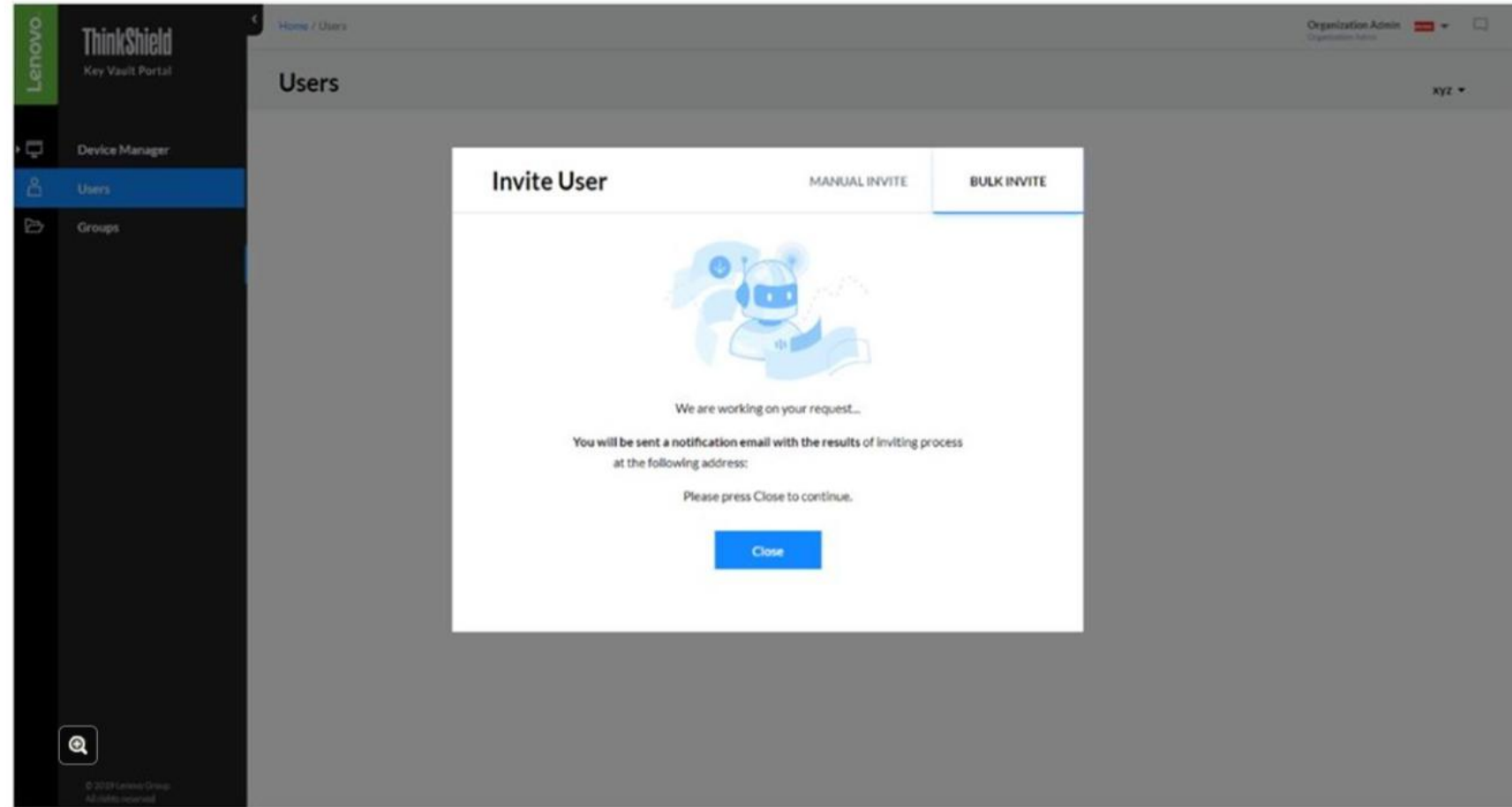


Step



ThinkShield Web application – Bulk Invite users

When the verification process is over, a confirmation email will be sent to the email address associated with your profile.

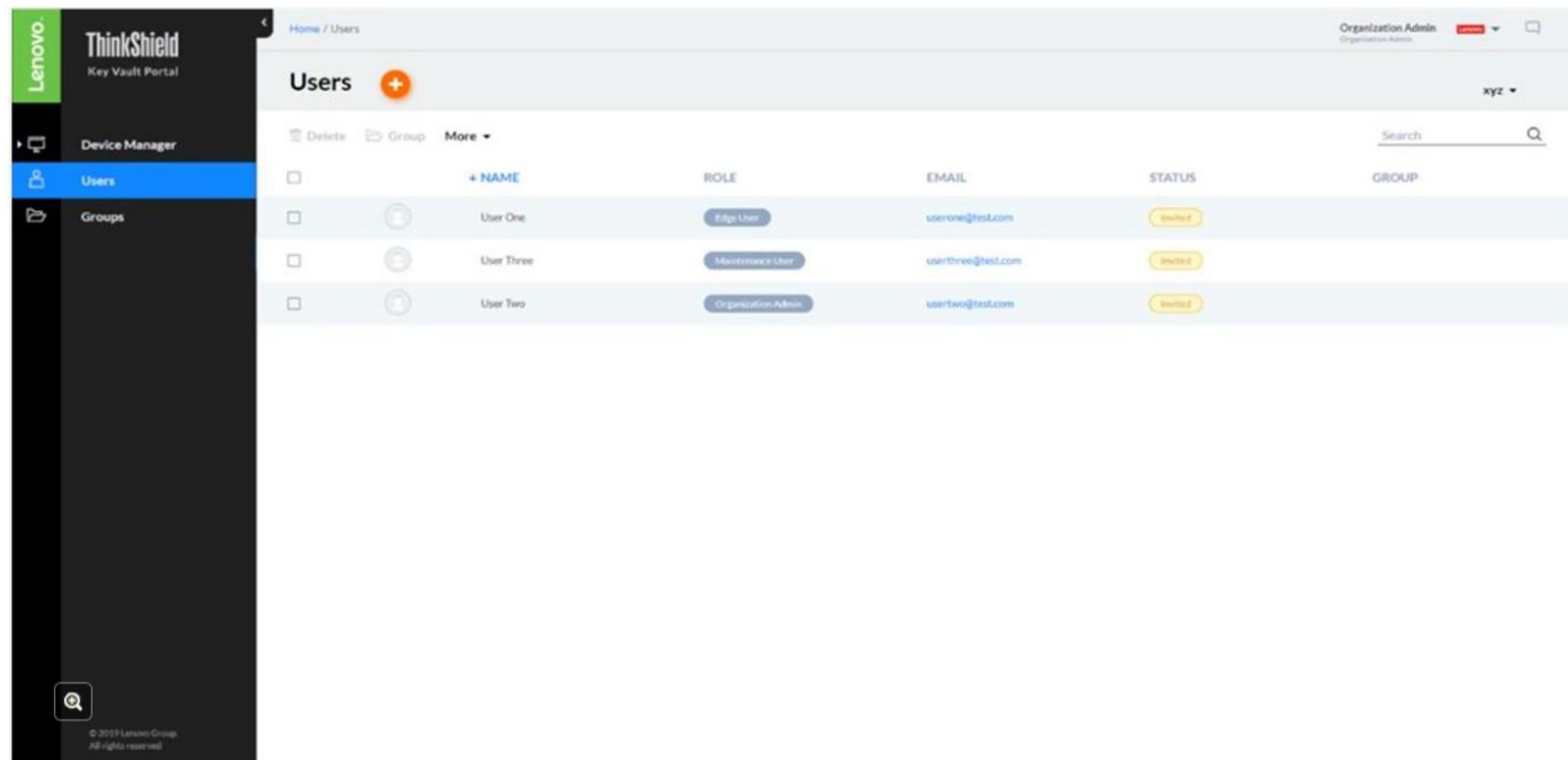


Step



ThinkShield Web application – Bulk Invite users

If the upload is successful, the users will be displayed in the **Users** list. The status of the users will be **Invited**.



The screenshot shows the ThinkShield Key Vault Portal interface. On the left is a dark sidebar with the Lenovo logo and navigation options: Device Manager, Users (highlighted), and Groups. The main content area is titled 'Users' and shows a table of three users. Each user row includes a checkbox, a user icon, a name, a role button, an email address, a status button labeled 'Invited', and a group name. The top right of the interface shows the user is logged in as 'Organization Admin'.

		+ NAME	ROLE	EMAIL	STATUS	GROUP
☐		User One	<button>Edge User</button>	userone@test.com	<button>Invited</button>	
☐		User Three	<button>Maintenance User</button>	userthree@test.com	<button>Invited</button>	
☐		User Two	<button>Organization Admin</button>	usertwo@test.com	<button>Invited</button>	

Step

1

2

3

4

5

